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NextGenYield

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Abstract

Well-timed detection of crop developmental stages is essential for improving agriculture management practices and reducing potential yield losses. This study proposes an intelligent crop monitoring system and decision support system utilizing the FoMO4Wheat agriculture, a self supervised model trained on extensive wheat imagery. A classification layer is added to the frozen transformer features to divide wheat canopy imagery into six key growth stages, from tillering to complete growth. These sequential observations are recorded into a temporal growth timeline, which is then used by a rule-based argonomic system to provide guidance on scheduling , nutrient application, and possible crop risks. By combining visual crop analysis with practical farming recommendations , the proposed system provides an easy to use , data driven tool for precision crop management.

Executive Summary

The executive summary should be one to two pages' of an overview of the information contained in the FYP report. It should give the reader an easy reference, in a very brief form, to the important information contained in the report and explained in more detail in the body of the report. People reading the report will use this section as a reference during presentations.

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Chapter 1 Introduction

In traditional farming wheat production depends upon visual observation and fixed schedule dates. However, field decisions require knowing the exact growth stage of the crop, which does not always follow a calendar. These methods may not consider changes in the temperature ,weather and the growth of crops which can have farmers giving wrong inputs at the wrong time ,this results in wasting the resources and reduces crop production.

To solve this problem, this project develops an intelligent crop monitoring system which is NextGenYield. It is based on FoMo4Wheat, a vision model trained on nearly 2.5 million wheat canopy pictures. This system identifies 6 important wheat growth stages using a classification layer. The photos are smartphone photos. It also tracks the growth of the crop and gives farmers simple recommendations for fertilizing and usage of water percentage at the right time with preventing crop hazards.

1.1 Purpose of this Document

The main purpose of this document is to give the details of the development and evaluation of our Final year project, NextGenYield. The document mentions the technological methodology,data pipeline, and field validation needed to turn advanced computer vision models into a reliable smartphone app for farmers. The document answers the question:

Can a wheat vision model which is trained using research images , accurately detect the wheat growth stages from ordinary photos taken in real farm condition?

Also it takes into consideration whether adding a simple classification layer can improve the model and allow it to be more useful in terms of giving useful recommendations to farmers. This report includes the completion of the system, its methods ,architecture, testing ,implementation ,limitations and testing with the possible improvement of the future work.

1.2 Intended Audience

The main audience for this report are the panel reviewing this FYP, agriculture experts interested in crop monitoring and tech developers working on mobile apps for farmers and researchers interested in using image based models to study the crop growth.

1.3 Definitions, Acronyms, and Abbreviations

FYP: Final Year Project

AI: Artificial Intelligence

ML: Machine Learning

DL: Deep Learning

ViT: Vision Transformer

FoMo4Wheat: Foundation Model for Wheat

WGSP: Wheat Growth Stage Prediction

API: Application Programming Interface

REST: Representational State Transfer

UI: User Interface

UX: User Experience

MVP: Minimum Viable Product

1.4 Conclusion

In Conclusion, ”NextGenYeild” is ready to transform precision agriculture by fusing a user-friendly mobile interface with a reliable AI-driven back-end system. In addition to streamlining crop monitoring for farmers, the platform makes use of cutting-edge technology like the FoMo4Wheat vision foundation model and downstream phenological classification to improve user experience and guarantee data validity. The team is dedicated to creating a platform that strikes a balance between computational accuracy and real-world field usability, ensuring that every stakeholder benefits from its services. This will be accomplished through thorough experimental evaluation on local field imagery and by grounding all decision-support in established agronomic principles.

(Note: This conclusion paragraph is a placeholder and will be updated to reflect the full chapter-by-chapter report outline once all subsequent chapters are finalized).

Chapter 2 Project Vision

Traditional farmers check their crops by checking them manually and with fixed date schedules. But due to the recent inventions in AI and computer vision has provided us with new ways to make farming easy and smarter. Our project aims to use these technologies to make a smart and much more intelligent crop monitoring system.This system will provide us with a strong foundation which can later be used to support different crops and various useful decision making for farmers. The aim of this chapter is to address the agriculture problems.

2.1 Problem Domain Overview

Wheat is the most needed and powerful crop in the world.it is a part of a major source of calories and protein for people all around the world. Therefore, wheat farming is very important for farming countries. During its growth it passes through several stages like stem elongation,booting and tillering. Each stage is in different needs of soil,water and nutrients. Changes in weather,seed and temperature affect the growth rate of wheat. And due to this farmers cannot rely on fixed dates to determine the stage of wheat. Therefore, regular monitoring is very important for the production of the yield crop.

2.2 Problem Statement

Improving farming practices for wheat yield needs sufficient water, fertilizer, and necessary treatment at accurate biological growth stages. However, farmers conventionally rely on visual checks or fixed calendar days, which results in growth variations due to weather and often leads to resource waste and less crop growth. Furthermore, existing solutions lack advance vision capabilities , unable to track temporal over time, reliable and evidence based guidance.

2.3 Problem Elaboration

Smart crop monitoring only works in real fields if one big farming problem and a few technical ones are solved. The main problem is that farmers give water and fertilizers at the wrong time because they cannot detect the growth stage of the wheat. On the technical side, phone cameras struggle outdoors because sunlight changes through the day, pictures are taken at awkward angles, and the soil in the background is cluttered, so older CNN models get confused. Stages that come one after another also look almost the same, and separating them requires noticing more than 700 tiny details of plant form that standard vision models cannot catch.

There is also very little labeled phone imagery from real local farms, so models have too few examples to train on. Finally, existing tools judge each picture by itself and do not keep a record of growth over time, which means they cannot tell farmers when to irrigate, fertilize, or protect their crops.

2.4 Goals and Objectives

Write goals and objectives here.

2.5 Project Scope

The scope of this project is a mobile application for wheat farmers,which monitors the growth stages of the crop and gives timely suggestions and decision support.

The application will allow farmers to:

- Take wheat canopy images and upload to the app using ordinary camera of the smartphone
- Classify the six important stages of wheat's growth which includes stages of (tillering, jointing, booting, anthesis, grain filling, and maturity)
- Visualise and track growth timelines for each field across the chronological observations.
- Receive rule-based and deterministic suggestions for irrigation timing, fertilizer scheduling, and hazard prevention.

For the Ai inference pipeline Python and PyTorch will be used. A pretrained vision model will be used as a foundation and a special layer for classification will be trained on top of it. This will be combined with a mobile framework such as React Native or Flutter for the frontend. The application will use REST API to communicate with the backend. Agile development principles will be used while developing the project and management of the project will be done iteratively.

The project deliverables will include:

- A functional mobile application and backend service for wheat growth-stage monitoring and farmer decision support
- A trained and validated downstream classification pipeline evaluated on realistic field imagery
- Comprehensive documentation of the design, implementation, and experimental testing

process

The project scope does not include the following:

- Utilising of autonomous drone or automated robotic hardware
- Estimating direct crop yield in absolute quantities (such as kilograms per hectare)
- Using Large Language Models to suggest unverified chemical dosage
- Use, deployment and maintenance of IoT sensors network.

2.6 Sustainable Development Goal (SDG)

NextGenYeild demonstrates how the platform contributes to sustainable agriculture as it improves crop productivity and helps in reliable food production. Ultimately, the project's goal is to strengthen harvest resilience and promote food security by helping ensure access to healthy and fresh food.



Figure 2.1: Sustainable Development Goal
This figure represents our sustainable development goal which is Zero Hunger

2.7 Constraints

The following are the main constraints of the system's design and implementation:

- Functional Scope: The image will strictly limit the functional scope to wheat growth staging with timeline tracking and decision support. No physical drones / IoT hardware nor direct yield forecasting.

- Compute Resources: Model adaptation and inference are limited to one of the local workstations of a member which has an Rtx 5070 Graphics card.
- Timeline: Each phase of development, testing and documentation must happen within 2 semesters (9-12 months).
- Team Allocation: In a group of three members, Engineering tasks, model experimentation and UI design are done in the project.
- Advisory Integrity: Recommendations should be given based on deterministic agronomic rules that are verified and validated. They must not come from a text generation model. AI techniques such as LLM or numerical AI can give wrong recommendations.
- Accessibility: The app should be lightweight , intuitive, and easily operable for farmers.

2.8 Business Opportunity

The potential for the app to commercially become a standalone business in the precision agriculture sector is very high. The business model can adopt a freemium approach providing basic growth-stage tracking for free while monetizing individual farmers by charging them their subscription fee for premium features. The premium features will include advanced timeline tracking, risk alerts, customised irrigation and fertiliser schedules specific to each field of a farmer. NextGen Yield can also be licensed as an enterprise B2B software or an API which can be used by agriculture corporations, fertilizer companies, and crop insurance firms. Through periodic visits and photographs, the organisation can monitor large areas of farmland that they have contracted.

2.9 Stakeholders Description/ User Characteristics

Identify and briefly describe who the users of this system will be, and their roles.

2.9.1 Stakeholders Summary

2.9.2 Key High-Level Goals and Problems of Stakeholders

Notice how a section's heading is changed to "heading 1". It will show up in the table of contents as a section under a chapter. Notice its numbering also. Similarly, subsections can have heading 2 and heading 3 styles.

NOTE: Subsections from 2.7 to 2.9 are optional for Research projects but compulsory for De-

velopment projects.

Chapter 3 Literature Review / Related Work

This chapter reviews related work that is already existing on crop monitoring, wheat growth stage detection and agricultural decision support systems that are related to NextGenYield. The literature is grouped into categories for example image-based wheat growth stage recognition, satellite and remote crop monitoring, image based plant disease detection, AI generated agronomic advice. By reviewing these systems we analysed their methods, capabilities, and limitations, while also highlighting gaps that provide a foundation for developing NextGenYeild as a practical smartphone-based crop monitoring and decision-support system.

3.1 Definitions, Acronyms, and Abbreviations

To be entered when this chapter is completed

3.2 Detailed Literature Review

In this section each related work is discussed in detail, moreover each work is critically analysed and explained how it relates or connects to NextGenYield.

3.2.1 Category A: Image-Based Wheat Growth-Stage Recognition

The studies in this category are closest to the core task of NextGenYield. They show that images can be used to detect growth stages of wheat but each of them depend on equipment, controlled conditions or sources of data that an ordinary farmer may not have access to.

3.2.1.1 Near-Surface Camera Phenology with Spatiotemporal Fusion

Summary

Cai et al. [1] developed a system that detects wheat growth stages by using fixed cameras. The system was at Baima Experimental Station of Nanjing Agricultural University in China. For 107 days a camera which was placed three metres above the ground took images from 8am to 5pm ,during the 2022-2022 growing season. The images were labeled into eight stages of growth ,from emergence to maturity. The researchers used ResNet50 as their base model and tried three different methods for combining spatial and chronological information . The best model was the one which combined an LSTM with ResNet50,and it achieved an accuracy of 93.5%. The study also found that results can be improved by taking higher resolution images, positioning the camera at about 40-60 degrees and fine tuning in two steps.

Critical Analysis of the Research Item

A strength of this study is that it carefully analyses design factors and confusion matrices that tell which stages are hard. The sequential model was weakest at anthesis. The finding that camera angle affects accuracy is useful for setting standard methods to capture image. However there are some practical limitations. The data was collected in one location only and in one growing season , moreover a fixed camera at a set height was used. Moreover the model required a sequence of 30 images instead of using one image. Since the series of images came from the same plots (6 total)and then split randomly, so performance was not demonstrated on an unseen field. The large model size also may need to be reduced for realtime use.

Relationship to the Proposed Research Work

nextGenYield handles the same task but it doesn't use installed cameras. Instead a farmer takes an image straight from his smartphone at irregular intervals, the classifier works on that single image, and the chronological reasoning and decisions are made within the app later on without any use of a neural network. The camera viewing angle technique helps us give farmers a guideline in the app of how to capture the image for best results.

3.2.1.2 GSP-AI: Growth Stage Prediction from Drone Images and Climate Data

Summary

Shen et al. [2] developed GSP-AI. It is a deep-learning platform that uses drone images and weather data to classify important wheat growth stages and predict flowering time. To train the modal they also created the Wheat Growth Stage Prediction (WGSP) dataset, which contains 70,410 labelled images from different wheat varieties in China, the United Kingdom, and the United States, along with related climate data. The study aims to help breeders and researchers in monitoring the growth stages of large numbers of varieties of wheat in different locations and different seasons.

Critical Analysis of the Research Item

The main strength is the dataset, it is large,consists of images of multiple countries, is openly available and is labelled by growth stage. This makes it a useful dataset for future. Combining images with climate data is also a good way to improve flowering-time prediction. However from a farmer's point of view there are some limitations. The system depends on drone flights and drone flights can be affected by rain[1].The model is designed for breeding research rather than helping farmers make decisions such as when to irrigate , fertilise etc. The paper also does

not provide any system that advises the farmer.

Relationship to the Proposed Research Work

WGSP provides the six-stage taxonomy and the dataset against which the NextGenYield classification head can be trained and/or evaluated. NextGenYield replaces drone imagery with smartphone photographs and for the classification step it does not require weather information. Moreover NextGenYield adds the missing layers absent in the study,which are a growth timeline and stage-specific advice.

3.2.1.3 DenseNet–BiLSTM Growth-Stage Determination from Image Sequences

Summary

Wang et al. [3] studied wheat growth-stage recognition . They growed wheat in a controlled climate room. Four different varieties of wheat were photographed every week from eighth fixed camera angles.The temperature of the room was kept at 23 °C with light for 16 hours and dark for 8 hoursand 90% humidity. The researchers removed the background using K-means colour clustering and used DenseNet201 model with a bidirectional LSTM to identify the five growth stages: tillering, re-greening, jointing, booting, and heading. The model achieved an accuracy of 98.43% on 40 test sequences.They performed better than basic CNN and CNN–LSTM models.

Critical Analysis of the Research Item

The study shows that results can be improved in detecting the growth stages if we use a sequence of images instead of a single image. However, the dataset is small, 592 sequences and 3483 images. The data is also unbalanced, with only 72 images showing the booting stage and only 7 of these in the test set. Booting had the lowest precision at 85.71% meaning the overall accuracy is a result of the outputs of the common stages. The authors also state that they faced problems in changing the lights, having parts of plants blocked, and addressing real field conditions before the system can be used in actual fields.

Relationship to the Proposed Research Work

This work is an example of why NextGenYield will report the metrics for each stage rather than one aggregate accuracy. It also highlights the gaps that NextGenYield aims to address by working with real field images that contain soil, shadows, nearby plants,weeds and an image that is not taken at a fixed angle or with a professional camera.

3.3 Literature Review Summary Table

The related work summary table is shown in 3.1

Table 3.1: Related Work Review Summary Table

This summary table lists the eight works discussed throughout this chapter.

Application	Features	Relevance	Limitations
Cai et al. [1]	Fixed-camera image series; CNN–LSTM fusion; eight stages	Shows camera angle and resolution affect stage accuracy	One station and one season; needs image series and fixed cameras; heavy model
GSP-AI [2]	Drone images plus climate data; WGSP dataset of 70,410 images	Source of the six-stage taxonomy and benchmark	Needs drone flights and weather records; built for breeding plots
Wang et al. [3]	DenseNet–BiLSTM on growth-chamber image sequences	Shows why per-class results matter (booting weakest)	Controlled environment; five stages; only 72 booting images
Bhatti et al. [4]	Sentinel-2 NDVI compared with daily field cameras, Sargodha, Punjab	Local evidence on the limits of satellite monitoring	Fog and cloud gaps; lower NDVI than cameras; no discrete stage labels
Mohanty et al. [5]	CNN on 54,306 PlantVillage leaf images	Illustrates the lab-to-field accuracy gap	99.35% on lab test set but 31.4% on images from other sources
Plantix [6]	Photo diagnosis of about 400 pests, diseases and deficiencies; treatment tips	Closest farmer-facing app, but disease-focused	Modest initial accuracy in independent test; no growth-stage tracking
Ibrahim et al. [7]	ChatGPT compared with extension agents on Nigerian rice questions	Evidence against free-text advice on rates and timing	Weak on planting time and fertiliser rate and timing; small sample
FoMo4Wheat [8]	ViT foundation model pretrained on 2.5 million wheat images	Backbone used by NextGenYield	Heavy for edge devices; wheat-focused; not trained on phone photos

3.4 Conclusion

Write the conclusion of the literature review. The columns in the table depending upon your problem and should be specific to your project as described in 3.1, ??, respectively. Proper citations and references are necessary. The template has already an integrated IEEE references format. Go to Google scholar, copy the bib text of your reference, and paste the reference into the fypbib.bib file. Then all you have to do is use [?].

NOTE: For development projects, describe related or similar work done by other teams and details of their methods/algorithms. For a research project, a detailed literature survey is expected.

Chapter 4 Software Requirement Specifications

This chapter provides a comprehensive and detailed specification of the software requirements for NextGenYield: Wheat Crop Decision Support and Vision AI System. It delineates the functional and non-functional requirements, defines quality attributes, hardware and software runtime constraints, user-interaction use cases, graphical user interface architecture, database entity-relationship models, data dictionaries, and an exhaustive risk analysis. The objective of this specification is to furnish an unambiguous engineering blueprint that enables researchers, software engineers, and agricultural domain specialists to understand, verify, and reproduce the entire software system.

4.1 List of Features

NextGenYield provides a mobile-first, AI-driven precision agriculture ecosystem tailored specifically for wheat smallholder farmers, landowners, agronomic professionals, and input manufacturers. The primary features of the system include:

1. Foundation Model-Powered Phenological Classification: Utilizes the specialized FoMo4Wheat Vision Transformer (ViT-B/14, 86.6M parameters) pre-trained on 2.5 million agricultural images. A downstream classification head classifies crop canopy imagery into six distinct biological growth stages (Tillering, Jointing, Booting, Flowering, Grain Filling, and Maturity) without requiring extensive computational resources on the client side.

2. Real-Time Edge Image Quality and Angle Compliance Guardrail: An onboard sensor-fusion filter running on the smartphone's client layer that leverages the hardware gyroscope, accelerometer, and Laplacian edge-variance computation. It prevents degraded or non-compliant field photography by actively guiding farmers into capturing oblique-to-downward canopy imagery ($45^\circ - 80^\circ$ tilt) with adequate sharpness before image transmission.

3. Decoupled Field Demarcation and GeoJSON Boundary Creation: An interactive GIS mapping tool allowing farmers and landowners to demarcate field parcel perimeters directly upon high-resolution satellite and cadastral map overlays. It provides four distinct boundary geometries—Square, Rectangle, Polygon (multi-vertex), and Circle (centre-pivot radial)—generating standardized RFC 7946 GeoJSON representations with real-time acreage and square-meter calculations.

4. Decoupled Crop Cycle Initialization: Land parcel boundary creation is strictly separated

from agricultural crop inputs. Once a field boundary is established, farmers initiate a 140-day crop cycle only when the land is prepared for sowing, entering seed variety (e.g., Akbar-2019, Dilkash-20), certified seed source, fertilizer manufacturer (FFC, Engro, Fatima), and sowing date exactly once.

5. Deterministic, Hallucination-Free Agronomic Advisory Engine: Eliminates probabilistic Large Language Model (LLM) text hallucinations by employing a rule-governed agronomic knowledge base. Crop growth stage transitions are deterministically mapped to verified provincial agricultural protocols, generating stage-specific irrigation schedules, fertilizer split dosages, and disease warnings (e.g., yellow stripe rust prevention during booting).

6. Temporal Crop Growth Timeline Engine: Maintains a persistent, timestamped chronological ledger of all visual observations, stage confidence scores, and photographic records per field. It tracks cumulative Days After Sowing (DAS) and computes phenological velocity to detect delayed germination or premature senescence.

7. Collective Farm Intelligence and Regional GIS Mapping: Provides a spatial visualization of neighboring wheat parcels across the agricultural district (e.g., Faisalabad division). Fields are color-coded by current phenological stage, enabling peer observation, regional water management coordination along canal waterways, and disease spread monitoring.

8. Peer Benchmarking and Proven Techniques Showcase: A collaborative peer showcase displaying verified top-yielding plots (54–62 Maunds/Acre). Farmers can inspect the complete 140-day crop cycle—including exact watering dates, weed filtering applications, fertilizer split timings, and seed variants—and adopt proven schedules directly into their own field calendar.

9. Farmer Accessibility, Bilingual Localization, and Audio Narration: Engineered specifically for varying levels of digital and textual literacy. The interface offers full Urdu and English bilingual toggles, high-contrast light mode optimized for harsh direct field sunlight, adjustable typography scaling, and integrated audio synthesis that reads agronomic advisories aloud in spoken Urdu.

4.2 Functional Requirements

The functional requirements specify the operational capabilities and behaviors that NextGenYield must support for its primary actors: Farmers/Landowners, Agricultural Stakeholders, and the Underlying System.

4.2.1 Functional Requirements for Farmers and Landowners

Farmers and landowners using the mobile application shall be able to:

1. Account Registration and Profile Authentication: Create a user profile using phone number authentication or credentials, storing personal details, primary village/district location, and language preference.
2. Demarcate Land Boundaries on GIS Map: Add and delineate farm parcels using interactive map boundary tools (Square, Rectangle, Multi-Vertex Polygon, and Centre-Pivot Circle) over satellite or cadastral views.
3. Inspect Calculated Area in Real Time: View automatically calculated parcel measurements in both standard agricultural Acres and metric square meters (m^2).
4. Initiate Crop Cycle: Activate an unstarted (fallow) field parcel by entering the wheat seed cultivar, certified seed supplier, fertilizer manufacturing brand, sowing technique, and sowing date.
5. Execute Sensor-Guided Field Photography: Capture canopy imagery using a camera interface equipped with real-time pitch angle reticles ($45^\circ - 80^\circ$ compliance) and Laplacian blur detection.
6. Receive Automated Growth Stage Classification: Obtain instant phenological identification across six wheat growth stages accompanied by an AI classification confidence score.
7. Access Actionable Agronomic Guidance: Retrieve deterministic recommendations detailing immediate irrigation requirements, fertilizer split quantities per acre, and active pathogen hazards.
8. Listen to Spoken Advisory Narration: Trigger text-to-speech audio guidance in spoken Urdu to comprehend recommendations without reading text on screen.
9. Review Chronological Field Timeline: Inspect sequential photographic and stage observations logged since sowing date to evaluate crop development pace.
10. Explore Regional GIS Map: View surrounding agricultural plots color-coded by growth stage and inspect public cadastral and agronomic attributes of neighboring fields.
11. Adopt Best-Practice Crop Schedules: Browse top-performing benchmark yields, examine full 140-day crop management cycles, and replicate successful schedules into their own active fields.

4.2.2 Functional Requirements for Agricultural Stakeholders

Agricultural extension officers, agronomic researchers, and fertilizer technical representatives shall be able to:

1. Monitor Regional Phenological Trends: Access aggregated geospatial analytics illustrating district-wide stage distributions (e.g., percentage of fields in Jointing vs. Booting).
2. Audit Verified Harvest Yields: Verify and record final weighbridge harvest yields (Maunds/Acre) for benchmark fields to maintain showcase integrity.
3. Update Deterministic Agronomic Protocols: Modify stage-dependent threshold parameters, fertilizer dosing recommendations, and fungicide alerts according to updated agricultural research directives.
4. Assess Input Performance Analytics: Evaluate comparative cultivar performance and fertilizer brand efficacy across varying soil textures (e.g., Sandy Loam vs. Clay Loam).

4.2.3 Functional Requirements of the NextGenYield System

The core NextGenYield system shall be capable of:

1. Edge Sensor Fusion and Guardrail Verification: Continuously evaluate hardware orientation vectors and compute frame variance to reject blurry or improperly oriented photographs prior to server upload.
2. Vision Foundation Model Inference: Resize input canopy images to 518×518 resolution, execute forward-pass feature extraction across frozen ViT-B/14 transformer layers, and evaluate linear classification logits.
3. Geospatial Coordinate Processing: Validate, project, and store field polygon geometries compliant with RFC 7946 GeoJSON specifications, executing spatial containment queries for district-wide mapping.
4. Deterministic Rule-Engine Resolution: Map current classified growth stage, cumulative days after sowing, and registered fertilizer manufacturer to deterministic advisory guidelines without generative model hallucination.
5. Temporal State Management: Dynamically track active vs. fallow field states, calculate developmental velocity, and trigger critical milestone alerts.
6. Bilingual Data Serialization: Serve internationalized string tables and synthesize Urdu audio strings for low-literacy accessibility.

4.3 Quality Attributes

To guarantee operational viability across rural agricultural environments, NextGenYield satisfies the following quality attributes:

- Usability and Ergonomics: The application interface utilizes high-contrast visual tokens, prominent icon-driven navigation, localized Urdu terminology, and single-tap audio narration, accommodating farmers with limited smartphone proficiency.
- Reliability: The system maintains high service availability during peak agricultural seasons (November wheat sowing through April harvest), ensuring advisory continuity.
- Performance and Latency: On-device edge guardrails provide instant angle feedback ($<$ 50 ms), and end-to-end vision model inference executes within 1.8 seconds over standard 3G/4G cellular connections.
- Agronomic Safety and Accuracy: Deterministic rule matching prevents over-fertilization or premature watering, eliminating toxic agricultural hazards.
- Modularity and Reusability: Client-side view controllers, edge sensors, backend microservices, and GIS engines are strictly decoupled through well-defined REST API interfaces.
- Scalability: Stateless model inference containers and spatial database indexing support thousands of concurrent field queries during peak morning and evening farming hours.

4.4 Non-Functional Requirements

4.4.1 Performance

- Edge-based sensor angle validation and blur computation must execute in real time on mobile hardware ($<$ 50 ms latency).
- AI growth stage inference through the backend vision service must complete within 2.0 seconds under a nominal 4G network connection.
- The backend REST API must sustain at least 500 concurrent user requests without response degradation exceeding 1.5 seconds.
- Regional GIS map rendering of 100 localized plot polygons must render within 1.2 seconds of viewport settlement.

4.4.2 Reliability and Availability

- The system shall achieve an operational availability of at least 99.5% throughout the 140-day wheat cultivation season.
- In the event of network disconnection, the mobile client must cache captured observations, field boundaries, and advisory cards locally, synchronizing automatically when connectivity resumes.

4.4.3 Usability and Accessibility

- All user-facing screens must support instantaneous toggling between English and localized Urdu.
- High-contrast light mode styling must enforce a contrast ratio exceeding $4.5:1$ (WCAG AA standard) to preserve legibility under direct ambient field sunlight exceeding 50,000 lux.
- All key agronomic recommendations must feature an audible spoken Urdu audio narration button accessible via a single touch.
- Typography must support standard ($1.0\times$) and enlarged ($1.15\times$) font multiplication modes.

4.4.4 Reusability and Maintainability

- The mobile application code base must follow a modular component architecture (under cross-platform frameworks such as React Native or Flutter), strictly isolating UI components, translation keys, state providers, and mock fixtures.
- The deterministic advisory logic must be maintained in structured configuration schemas rather than hardcoded conditional trees, permitting rapid agronomic parameter updates without recompiling client binaries.

4.4.5 Security and Data Integrity

- All telemetric observations, user credentials, and spatial coordinates transmitted over the public network must be encrypted via Transport Layer Security (TLS 1.3).
- Field polygon definitions must enforce strict topological validation (no self-intersecting loops or degenerate nodes) before persisting to the spatial database.

4.5 Assumptions

- Hardware Availability: The primary user possesses an Android or iOS smartphone equipped with a functional optical camera, accelerometer, gyroscope, and GNSS/GPS location services.
- Cellular Connectivity: While the client supports local caching for offline observation capture, periodic internet connectivity (2G/3G/4G or Wi-Fi) is assumed for model inference synchronization and satellite basemap tile streaming.
- Crop Scope: The current operational scope of NextGenYield focuses exclusively on bread wheat (Triticum aestivum) cultivated within irrigated sub-tropical agricultural regimes (e.g., Indus Basin plains).
- Third-Party Basemaps: Map rendering depends on standard OpenStreetMap or Mapbox satellite raster tile services for aerial spatial boundary visualization.

4.6 Hardware and Software Requirements

4.6.1 Hardware Requirements

- Development Workstation:
  - Multi-core processor (Intel Core i7/AMD Ryzen 7 or Apple Silicon M-series).
  - 16 GB DDR4/DDR5 RAM.
  - 50 GB free solid-state storage.
  - Dedicated GPU (NVIDIA RTX 3060 or higher with 8 GB+ VRAM) for model fine-tuning and PyTorch inference benchmarking.
- Target Client Mobile Device:
  - Operating System: Android 10.0+ (API level 29+) or iOS 15.0+.
  - Camera: Minimum 8.0-megapixel rear optical sensor with autofocus.
  - Integrated Sensors: 3-axis accelerometer and 3-axis gyroscope.
  - Minimum RAM: 2.0 GB.
  - Network: 3G, 4G LTE, or Wi-Fi radio.

4.6.2 Software Requirements

- Mobile Client Runtime: Modern cross-platform mobile application framework (React Native or Flutter) supporting native hardware sensor access (accelerometer, gyroscope, camera), reactive state management, and integrated GIS mapping canvas.
- Programming Languages: TypeScript / Dart (client-side mobile application), Python 3.10+ (backend vision API and deterministic advisory services).
- AI/Vision Framework: PyTorch 2.x, Hugging Face Transformers, TorchVision, FoMo4Wheat ViT-B/14 pre-trained model checkpoint.
- Backend Web Services: FastAPI / Python ASGI runtime, Uvicorn server, Pydantic data schemas.
- Database Management System: PostgreSQL 15+ equipped with PostGIS extension for spatial polygon geometries and JSONB support for temporal cycle logs.
- Operating Systems: Windows 11 / Linux (Ubuntu 22.04 LTS) for development, target deployment on Android and iOS.

4.7 Use Cases

NextGenYield's operational workflow is structured into nine core use cases that cover user authentication, land demarcation, cycle initialization, AI canopy phenotyping, advisory generation, timeline tracking, GIS exploration, and peer benchmark adoption. Each usecase table details the actor actions and corresponding system responses across standard and alternative execution paths.

Table 4.1: Use Case 1: User Registration

Name	User Registration		
Actors	Smallholder Farmer, Landowner, Agricultural Stakeholder		
Summary	A new user creates an account within the NextGenYield platform to manage farm parcels and receive personalized crop advisories.		
Pre-Conditions	The user has installed the NextGenYield application and possesses an active mobile telephone number.		
Post-Conditions	A user profile is created in the database, and authentication tokens are securely saved to the device.		
Special Requirements	The user interface must support clear Urdu and English text prompts with large touch targets.		
Basic Flow			
Actor Action		System Response	
1	User launches the application and selects “Create Account”.	2	The system displays the registration form.
3	User enters full name, mobile number, district, and preferred language (Urdu or English).	4	System records input data and validates telephone format.
5	User creates a secure PIN or password.	6	System verifies PIN strength and confirms credentials match.
7	User submits the registration form.	8	System creates user record, stores session tokens, and navigates to the Farm Dashboard.
Alternative Flow			
3	User enters an already registered mobile number.	4-A	System responds with an error message: Phone number already registered, and redirects to Login.
7	User submits registration form without internet connectivity.	8-A	System responds with network failure alert and prompts user to retry.

Table 4.2: Use Case 1: User Registration

Name	User Registration		
Actors	Smallholder Farmer, Landowner, Agricultural Stakeholder		
Summary	A new user creates an account within the NextGenYield platform to manage farm parcels and receive personalized crop advisories.		
Pre-Conditions	The user has installed the NextGenYield application and possesses an active mobile telephone number.		
Post-Conditions	A user profile is created in the database, and authentication tokens are securely saved to the device.		
Special Requirements	The user interface must support clear Urdu and English text prompts with large touch targets.		
Basic Flow			
Actor Action		System Response	
1	User launches the application and selects 'Create Account'.	2	The system displays the registration form.
2	User enters full name, mobile number, district, and preferred language (Urdu or English).	3	System records input data and validates telephone format.
3	User creates a secure PIN or password.	4	System verifies PIN strength and confirms credentials match.
4	User submits the registration form.	5	System creates user record, stores session tokens, and navigates to the Farm Dashboard.
Alternative Flow			
2-A	User enters an already registered mobile number.	3-A	System responds with an error message: Phone number already registered, and redirects to Login.
4-A	User submits registration form without internet connectivity.	5-A	System responds with network failure alert and prompts user to retry.

Table 4.3: Use Case 2: User Login

Name	User Login		
Actors	Registered Farmer, Landowner, Stakeholder		
Summary	An existing user authenticates into the NextGenYield mobile app to access their fields, timelines, and advisories.		
Pre-Conditions	The user has an active account on the platform.		
Post-Conditions	User session is authenticated, and the personalized farm overview dashboard is rendered.		
Special Requirements	Supports rapid biometric or PIN entry for smallholder accessibility.		
Basic Flow			
Actor Action		System Response	
1	User opens the application.	2	The system displays the login interface requesting mobile number and PIN.
2	User enters registered mobile number and PIN.	3	System verifies credentials against the user database.
3	User taps the 'Sign In' button.	4	System initializes user session, loads farm parcel data, and navigates to Dashboard.
Alternative Flow			
2-A	User enters invalid mobile number or PIN.	3-A	System responds with an error message: Incorrect credentials entered, and prompts re-entry.
3-A	Device is offline during login attempt.	4-A	System validates locally cached authentication token to grant offline access.

Table 4.4: Use Case 3: Field Boundary Demarcation on Map (GeoJSON)

Name	Field Boundary Demarcation on Map (GeoJSON)		
Actors	Farmer, Landowner		
Summary	The user demarcates the physical perimeter of their agricultural parcel directly on a satellite map using interactive geometric tools.		
Pre-Conditions	The user is logged in and navigates to the Map tab.		
Post-Conditions	A new parcel record is saved with RFC 7946 GeoJSON boundary coordinates, calculated acreage, and unstarted (fallow) cycle status.		
Special Requirements	High-resolution satellite basemap overlay with real-time acreage calculation.		
Basic Flow			
Actor Action		System Response	
1	User taps 'Add Field' on the map interface.	2	System centers satellite view on user GPS coordinates and displays geometry selector tools.
2	User selects boundary tool: Square, Rectangle, Multi-vertex Polygon, or Circle.	3	System activates the interactive drawing canvas on the satellite view.
3	User taps or drags on satellite view to delineate parcel boundaries.	4	System computes and displays parcel acreage (Acres and m^2) in real time.
4	User enters custom field label (e.g., 'North Canal Plot') and confirms.	5	System serializes perimeter into RFC 7946 GeoJSON, saves parcel, and updates map.
Alternative Flow			
3-A	User creates a self-intersecting polygon loop.	4-A	System flags topological error and prompts user to adjust boundary vertices.
4-A	User cancels demarcation before confirming.	5-A	System discards temporary vertices and restores default map viewport.

Table 4.5: Use Case 4: Starting a Crop Cycle (Seed Variety & Fertilizer Configuration)

Name	Starting a Crop Cycle (Seed Variety & Fertilizer Configuration)		
Actors	Farmer		
Summary	The farmer activates a fallow field parcel by initializing a new 140-day wheat cultivation cycle, configuring seed cultivar, supplier, and fertilizer brand in a single entry step.		
Pre-Conditions	A field boundary exists with no active cycle (Status: 'Cycle Not Started').		
Post-Conditions	A new 'CropCycle' record is activated for the field, initial watering/nutrient schedules are generated, and the cycle clock (DAS) commences.		
Special Requirements	Single-entry modal ensuring seed and fertilizer partner details are configured once per season.		
Basic Flow			
Actor Action		System Response	
1	Farmer taps 'Start Cycle' on a fallow field card.	2	System presents the Single-Entry Cycle Initialization Modal.
2	Farmer selects sown Wheat Variety (e.g., Akbar-2019, Dilkash-20) and Certified Seed Source.	3	System records cultivar and certified seed parameters.
3	Farmer selects Fertilizer Partner (FFC, Engro, Fatima) and enters Sowing Date.	4	System computes target 140-day phenological schedule.
4	Farmer taps 'Confirm & Start Cycle'.	5	System updates field state to 'Active Cycle', logs Day 0, and displays baseline advisory.
Alternative Flow			
3-A	Farmer enters a future sowing date.	4-A	System responds with warning: Sowing date cannot be in the future, and restricts confirmation.
4-A	Farmer exits modal without confirming.	5-A	System retains field in 'Cycle Not Started' state with no cycle created.

Table 4.6: Use Case 5: Crop Canopy Scanning with Quality & Angle Guardrail

Name	Crop Canopy Scanning with Quality & Angle Guardrail		
Actors	Farmer		
Summary	The farmer uses the smartphone camera to capture a standardized canopy photograph, guided by on-device angle reticles and blur detection filters.		
Pre-Conditions	An active crop cycle is selected, and camera/motion sensor permissions are granted.		
Post-Conditions	A verified high-quality canopy image is transmitted to the FoMo4Wheat vision inference pipeline.		
Special Requirements	Real-time hardware gyroscope/accelerometer sensor fusion and Laplacian edge variance computation (< 50 ms).		
Basic Flow			
Actor Action		System Response	
1	Farmer taps 'Scan Field Canopy' on the active field card.	2	System opens camera viewfinder with pitch angle reticle and blur indicator.
2	Farmer tilts phone toward crop canopy at $45^\circ - 80^\circ$ downward angle.	3	System sensor listener confirms tilt is within optimal green compliance zone.
3	Farmer holds camera steady over the wheat foliage.	4	System computes Laplacian variance (> 90) and unlocks shutter button.
4	Farmer taps shutter button to capture canopy image.	5	System saves photograph locally and transmits it to vision inference backend.
Alternative Flow			
2-A	Camera is held flat or horizontally ($< 45^\circ$ or $> 80^\circ$).	3-A	System locks shutter and HUD reticle displays warning: 'Tilt downward toward crop canopy'.
3-A	Camera motion causes excessive image blur (< 90).	4-A	System locks shutter and displays alert: 'Hold camera steady'.

Table 4.7: Use Case 6: Retrieving Agronomic Advisory & Growth Guidance

Name	Retrieving Agronomic Advisory & Growth Guidance		
Actors	Farmer, Landowner		
Summary	The system provides verified, deterministic agronomic guidance based on the classified growth stage, DAS, and registered fertilizer partner.		
Pre-Conditions	A canopy observation has been classified by the FoMo4Wheat model.		
Post-Conditions	Recommended irrigation actions, fertilizer split requirements, and disease alerts are presented to the farmer.		
Special Requirements	Hallucination-free deterministic rule engine and one-tap spoken Urdu audio narration.		
Basic Flow			
Actor Action		System Response	
1	Backend vision service classifies the canopy photograph.	2	System returns classified growth stage (e.g., Booting Stage, 94.2% confidence).
2	Deterministic rule engine matches stage, DAS, and fertilizer partner.	3	System compiles stage-specific watering advice, fertilizer dosage, and rust warnings.
3	Farmer views the Stage Advisory Card on screen.	4	System highlights critical pending milestones and nutrient recommendations.
4	Farmer taps 'Listen (Urdu Audio)' speaker button.	5	System synthesizes and plays audio narration in spoken Urdu.
Alternative Flow			
4-A	Device audio is muted or volume is set to zero.	5-A	System displays on-screen toast: 'Please raise device volume to listen'.
2-A	Network timeout occurs during AI classification.	3-A	System serves cached baseline advisory for current DAS with offline disclaimer.

Table 4.8: Use Case 7: Field Observation Logging & Timeline Tracking

Name	Field Observation Logging & Timeline Tracking		
Actors	Farmer, Agronomic Extension Worker		
Summary	The user inspects the historical progression of the crop cycle from sowing to harvest through a sequential visual timeline.		
Pre-Conditions	An active crop cycle has at least one logged observation or milestone.		
Post-Conditions	The user reviews developmental milestones, phenological velocity, and photographic records across the 140-day lifecycle.		
Special Requirements	Chronological milestone ledger with photographic evidence and DAS progression.		
Basic Flow			
Actor Action		System Response	
1	User navigates to Field Details and selects 'Timeline' tab.	2	System retrieves chronological log of cycle events from database.
2	User scrolls through 140-day growth milestone nodes.	3	System renders DAS markers (Day 0 Sowing, Day 22 CRI, Day 50 Jointing, etc.).
3	User taps a specific historical observation node.	4	System expands node showing canopy photo, classified stage, and historical advisory.
4	User compares current growth stage against target schedule.	5	System displays phenological pacing indicator (Normal, Delayed, or Advanced).
Alternative Flow			
2-A	No scans have been logged yet for active cycle.	3-A	System displays placeholder message prompting user to perform initial scan.

Table 4.9: Use Case 8: Regional Farm Map Exploration & Neighbor Plot Inspection

Name	Regional Farm Map Exploration & Neighbor Plot Inspection		
Actors	Farmer, Landowner, Agricultural Stakeholder		
Summary	The user navigates a regional GIS map displaying their own parcels alongside neighboring plots, color-coded by crop growth stage.		
Pre-Conditions	GPS location services are active or a target district is selected.		
Post-Conditions	Regional spatial distribution of wheat phenology is visualized for community coordination and peer learning.		
Special Requirements	Stage-based polygon color coding and privacy-preserving public agronomic attributes.		
Basic Flow			
Actor Action		System Response	
1	User taps the 'Regional Map' tab.	2	System fetches spatial boundary polygons for the district and renders map tiles.
2	User views parcels color-coded by stage (Green for Tillering, Blue for Booting, Amber for Maturity).	3	System displays interactive plot overlays.
3	User taps an adjacent neighboring parcel polygon.	4	System opens bottom inspection sheet showing wheat cultivar, current stage, and field acreage.
4	User inspects regional canal water turn schedule.	5	System displays community water management notices.
Alternative Flow			
1-A	Device lacks internet connectivity.	2-A	System renders locally cached vector plot outlines with offline status banner.
3-A	User attempts to view private owner information.	4-A	System restricts personal data, displaying only anonymized agronomic attributes.

Table 4.10: Use Case 9: Adopting Proven Best-Practice Yield Schedule (Showcase)

Name		Adopting Proven Best-Practice Yield Schedule (Showcase)	
Actors		Farmer	
Summary		The farmer explores verified high-yielding benchmark fields (54–62 Maunds/Acre) in the Proven Techniques Showcase and adopts their 140-day management schedule.	
Pre-Conditions		The farmer has an active or planned wheat crop cycle.	
Post-Conditions		The benchmark field's watering and fertilizer schedule is replicated into the farmer's crop cycle calendar.	
Special Requirements		Complete 140-day event schedule display (watering days, weed filtering, fertilizer splits) and one-tap adoption.	
Basic Flow			
Actor Action		System Response	
1	Farmer taps 'Proven Techniques' tab.	2	System lists verified high-yield benchmark plots (54–62 Maunds/Acre).
2	Farmer selects benchmark plot (e.g., 'Samundri Plot A: 62 Maunds/Acre, Dilkash-20').	3	System displays full 140-day management event log.
3	Farmer reviews exact dates for watering, weed filtering, and fertilizer splits.	4	System highlights yield gain factors.
4	Farmer taps 'Adopt Schedule for My Field'.	5	System synchronizes schedule into farmer active cycle calendar and sets reminder alerts.
Alternative Flow			
4-A	Farmer has no active or planned crop cycle.	5-A	System prompts user to first add a field boundary and start a crop cycle.

4.8 Graphical User Interface

The NextGenYield mobile user interface is engineered to combine agricultural usability under direct sunlight with high-accuracy AI phenotyping tools. Below are the architectural layout specifications and figure placeholders for the system's graphical screens:

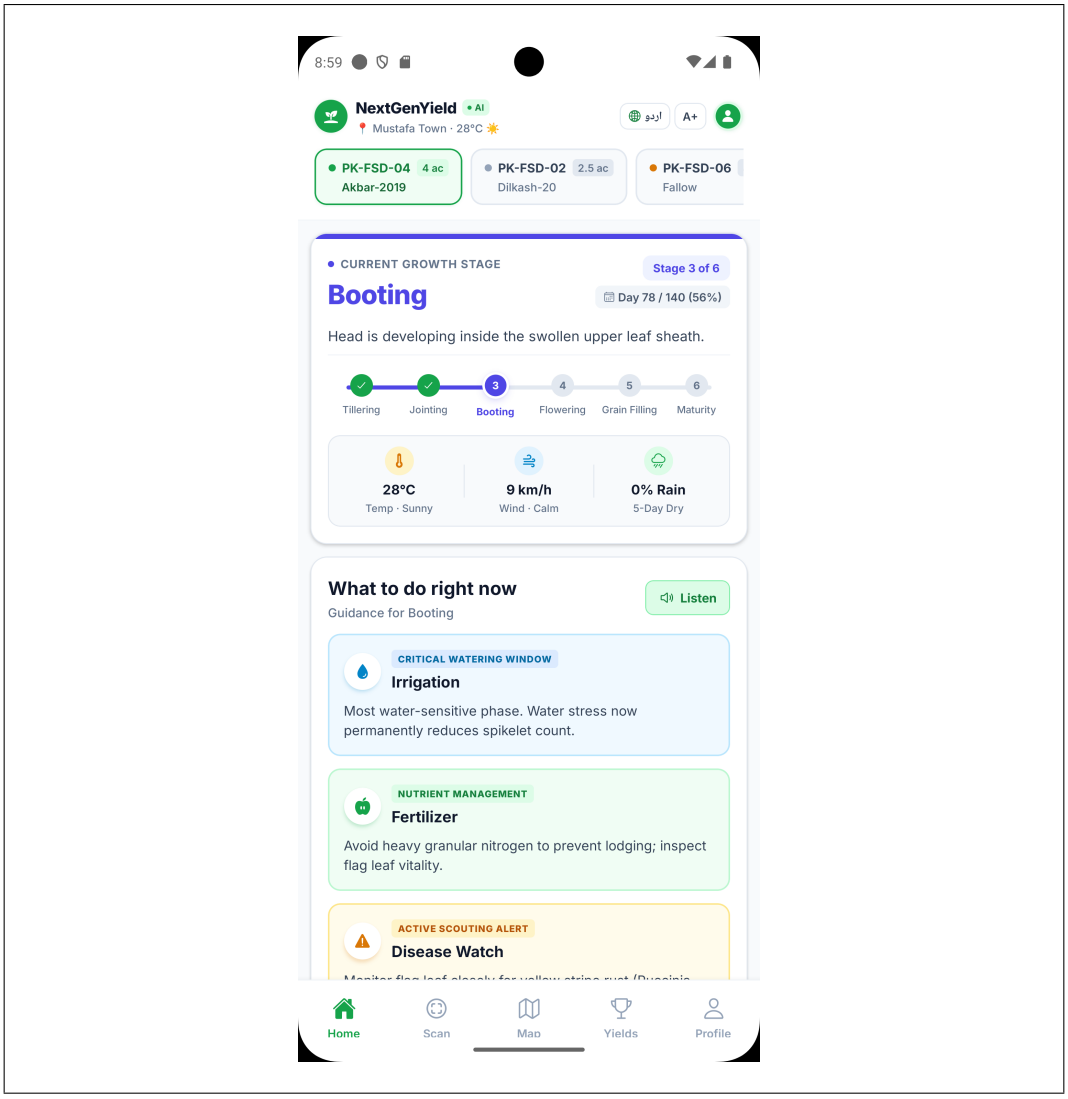


Figure 4.1: Farmer Home Dashboard and Cycle Management View

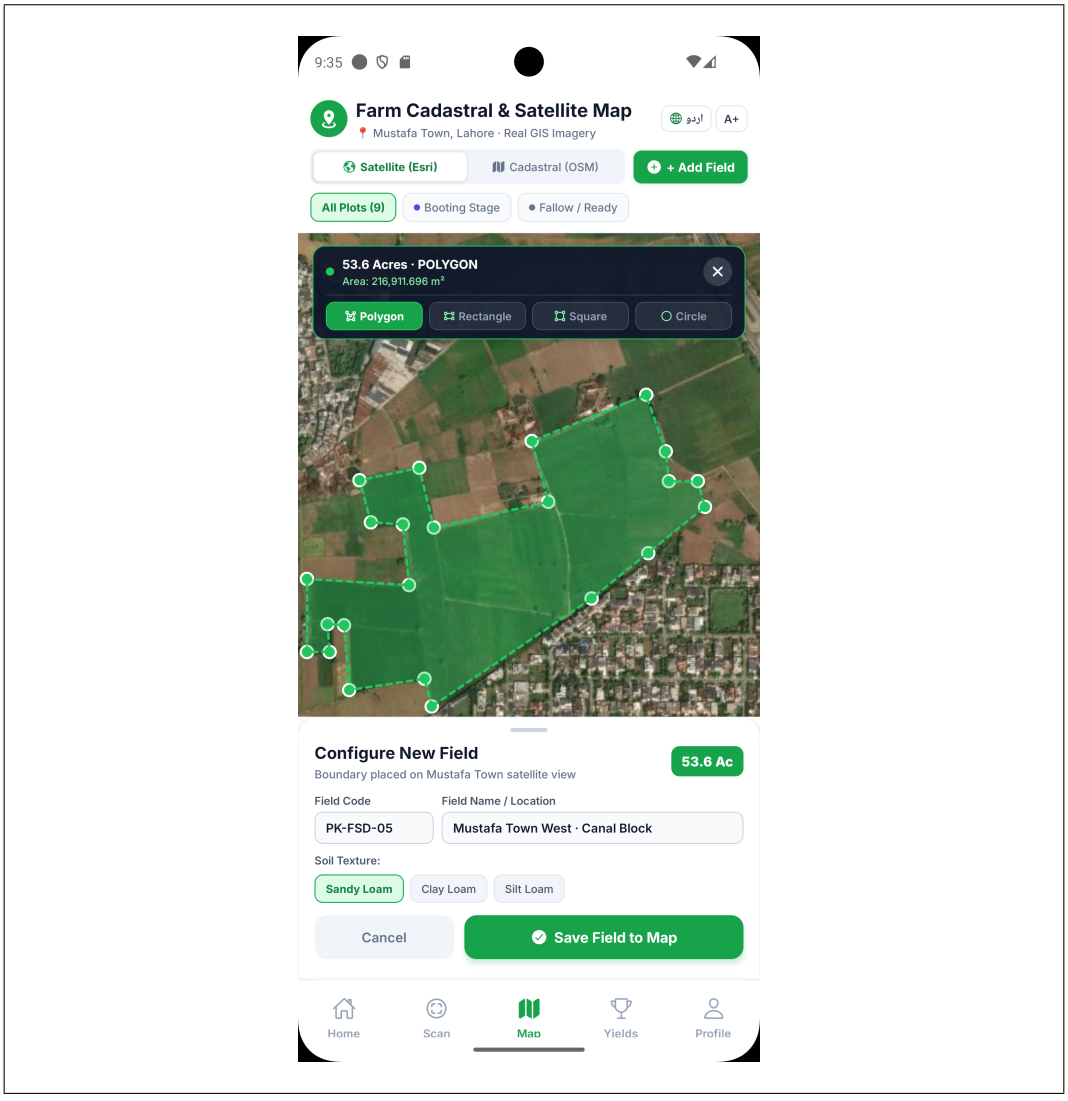


Figure 4.2: Decoupled Interactive GeoJSON Map Demarcation Interface

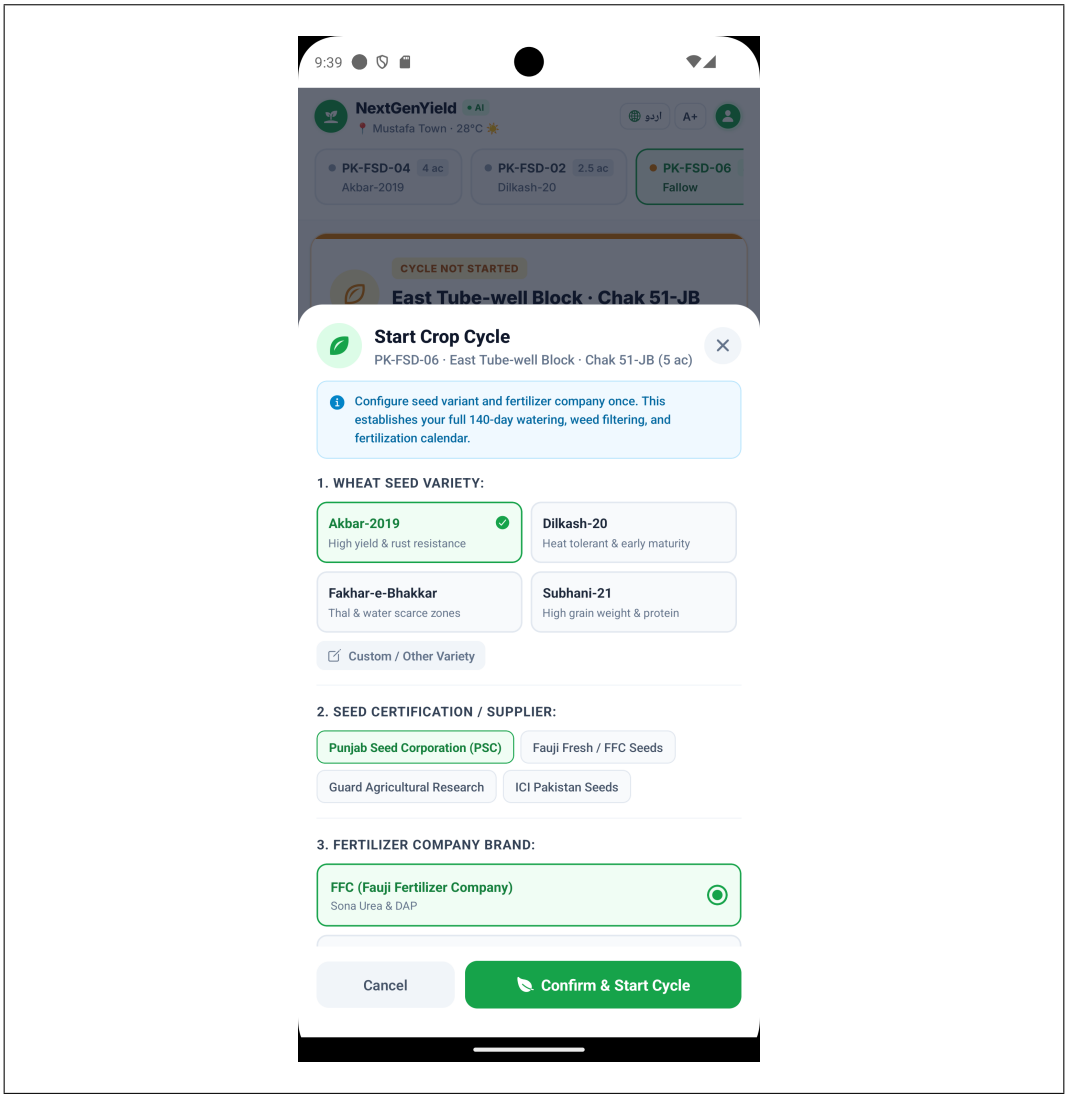


Figure 4.3: Single-Entry Crop Cycle Initialization Modal

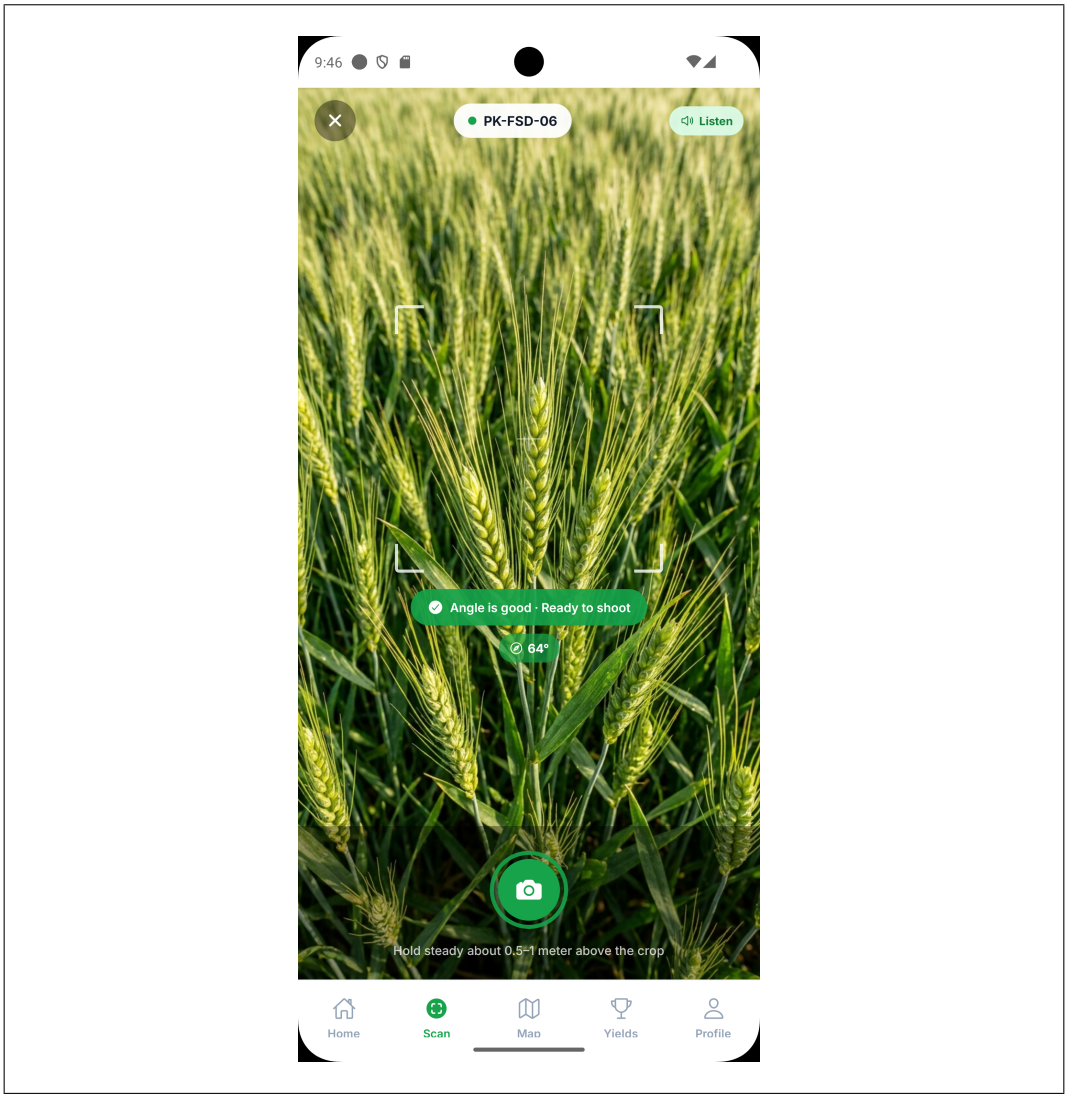


Figure 4.4: Modal guided Field Photography Guardrail

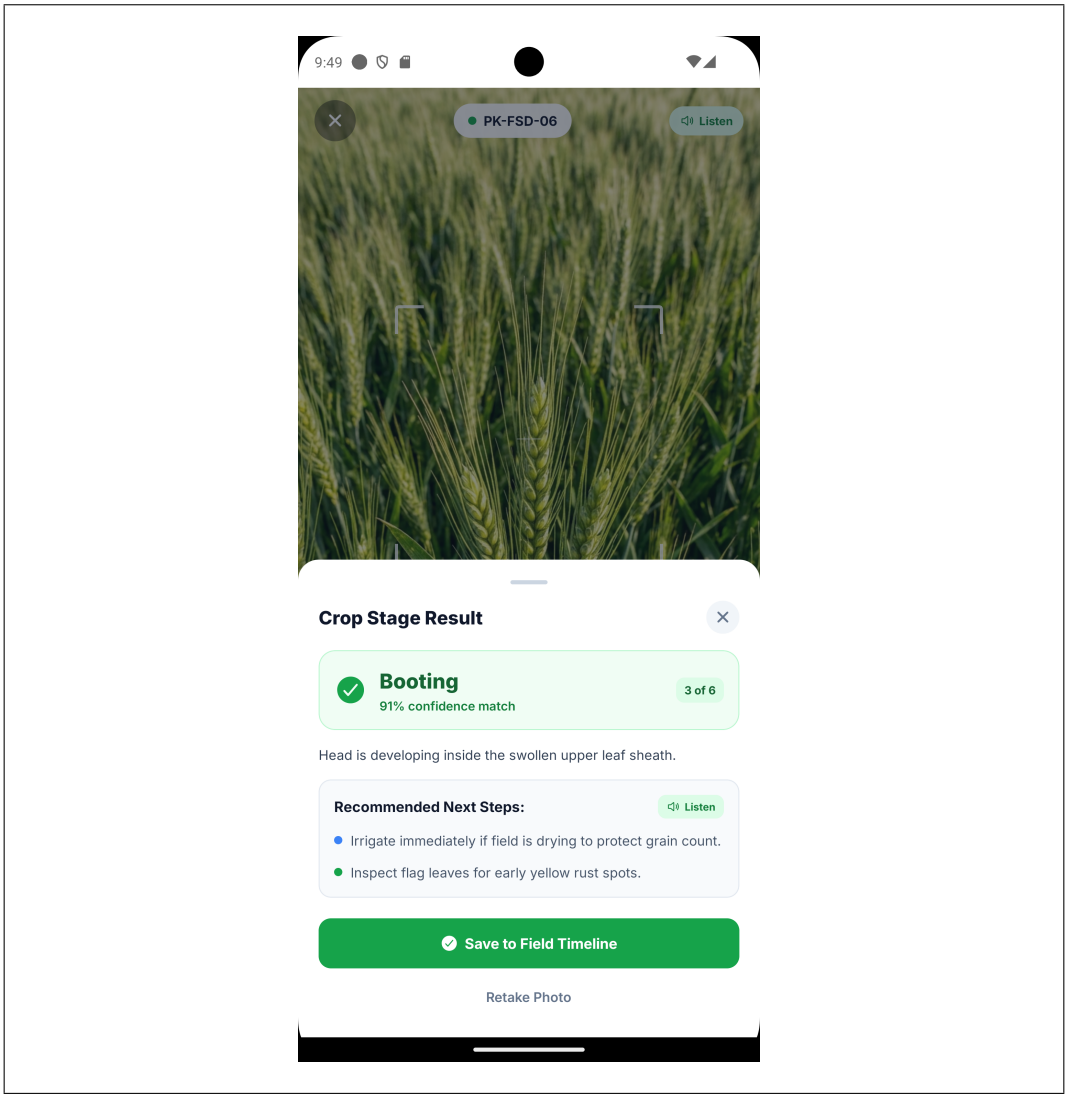


Figure 4.5: Phenological AI Classification and Agronomic Advisory Card

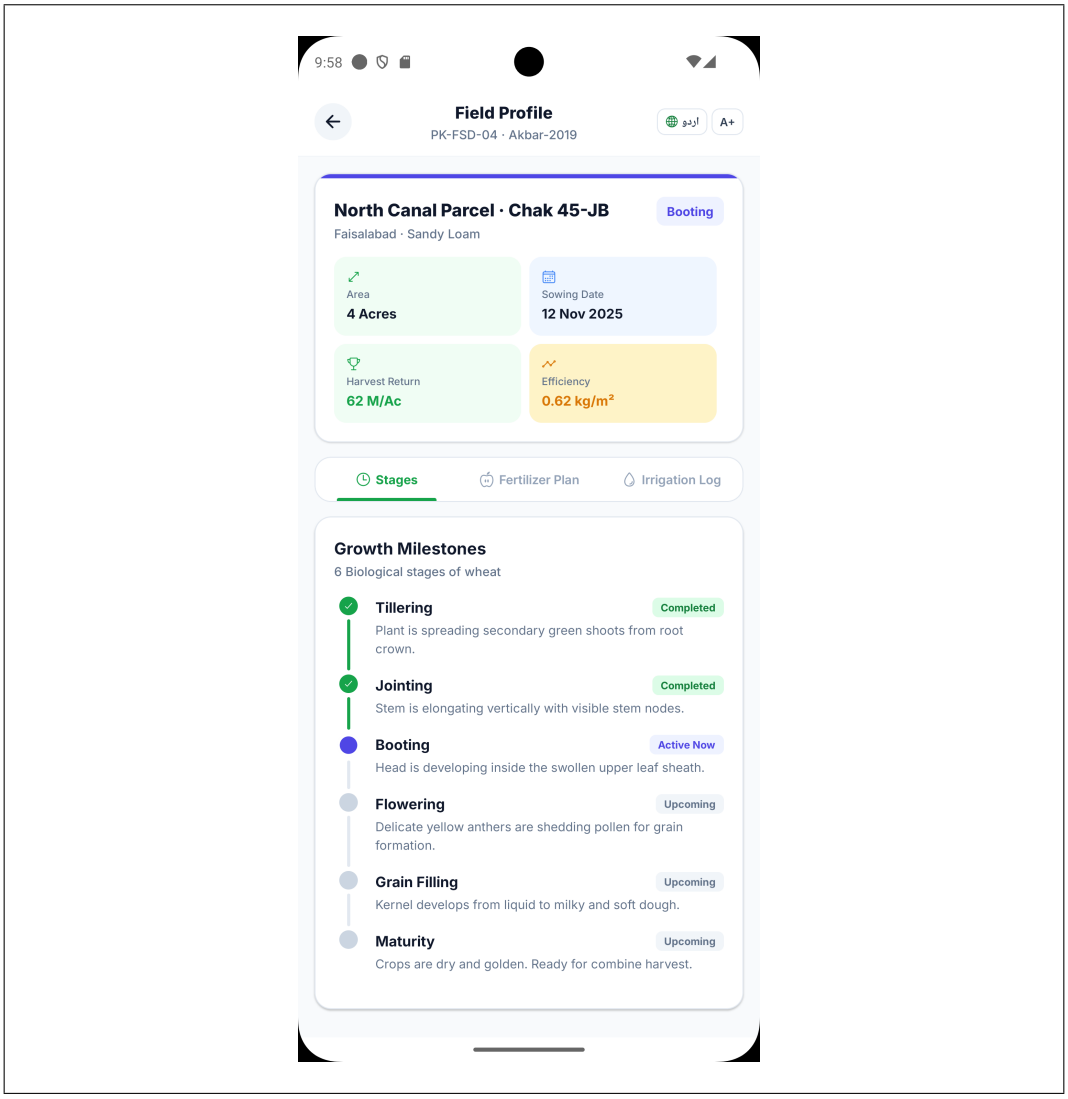


Figure 4.6: Temporal Crop Growth Timeline Screen

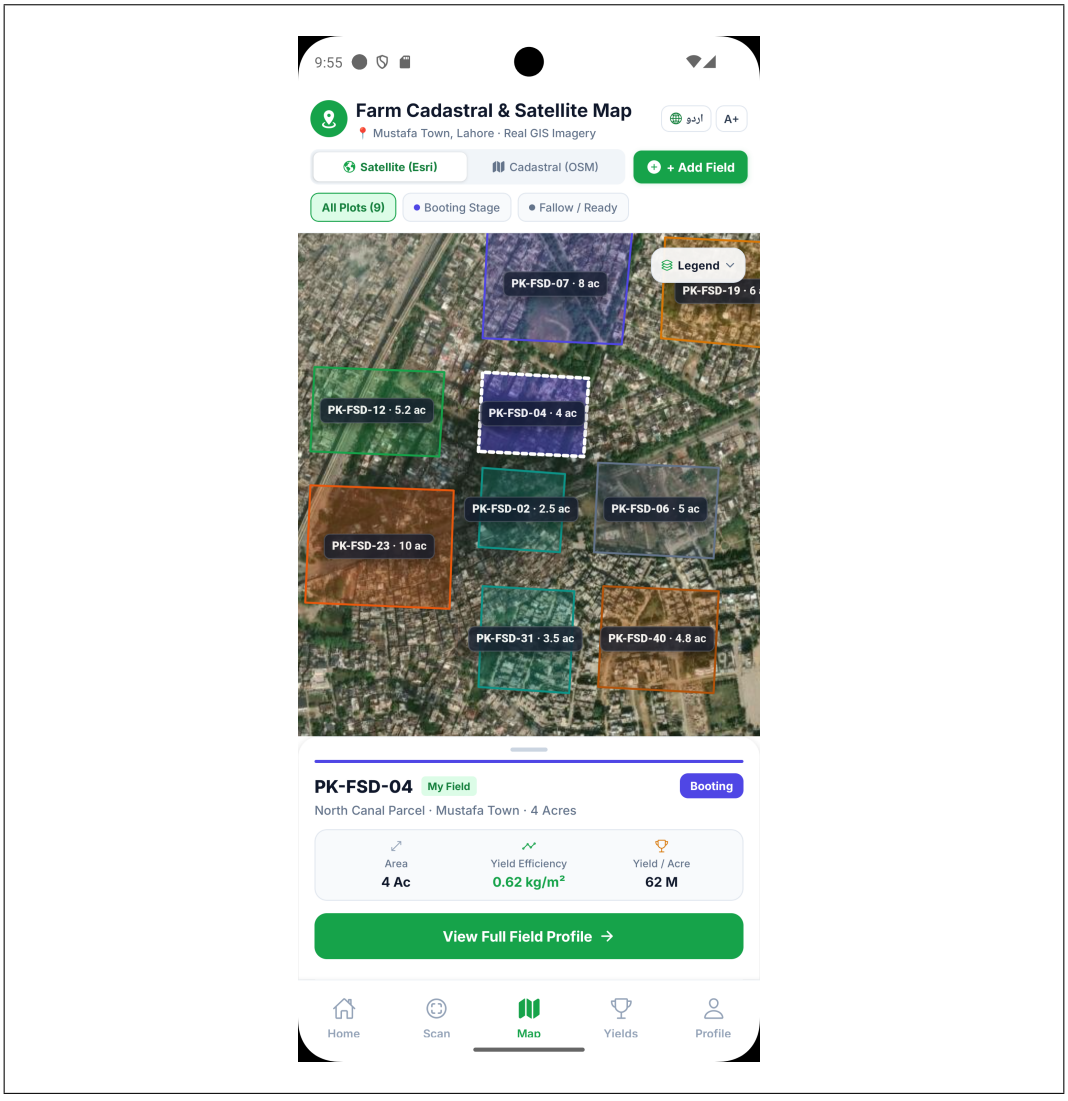


Figure 4.7: Regional GIS Farm Network and Neighbor Plot Inspection

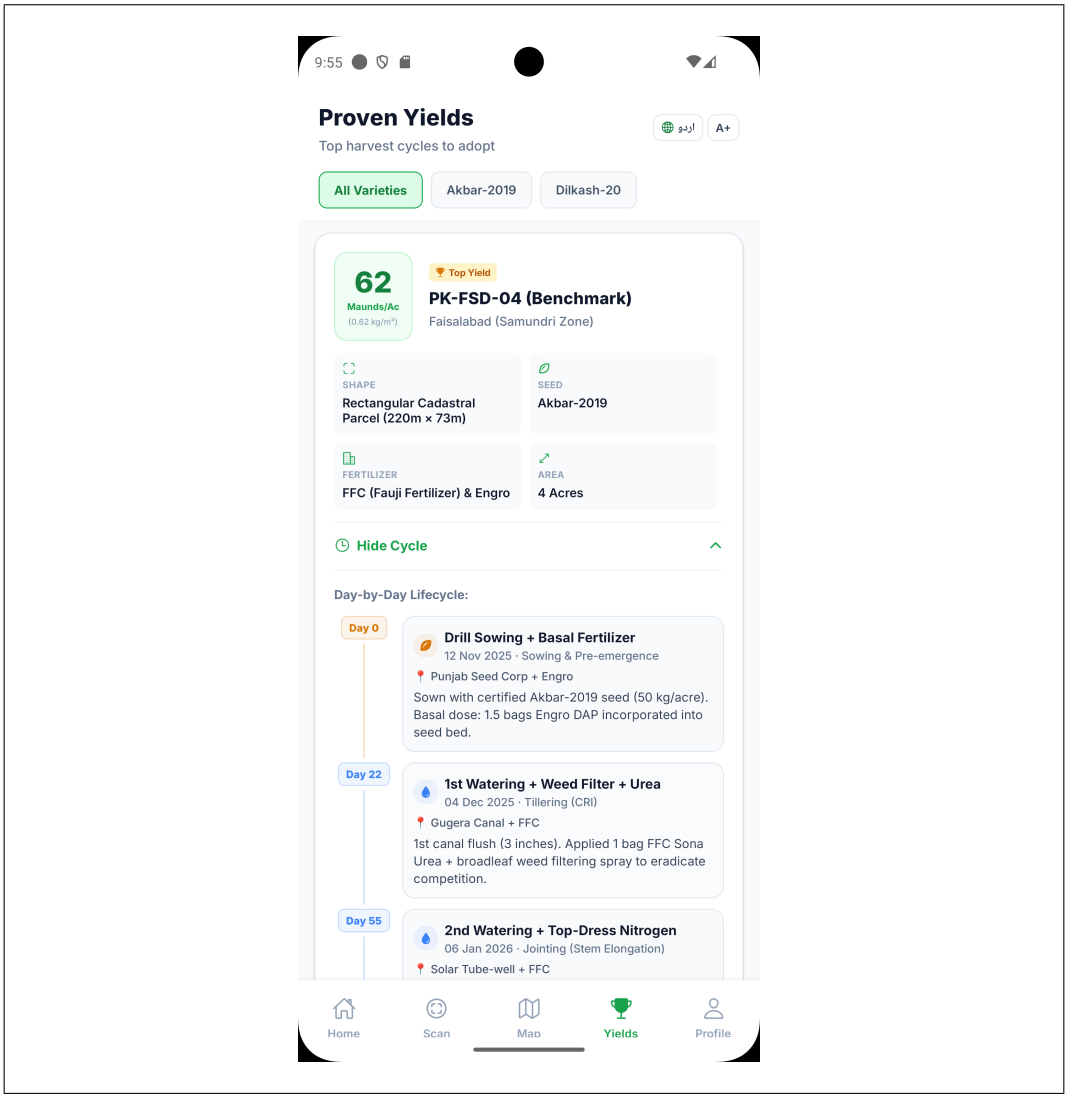


Figure 4.8: Proven Techniques Showcase and 140-Day Event Log

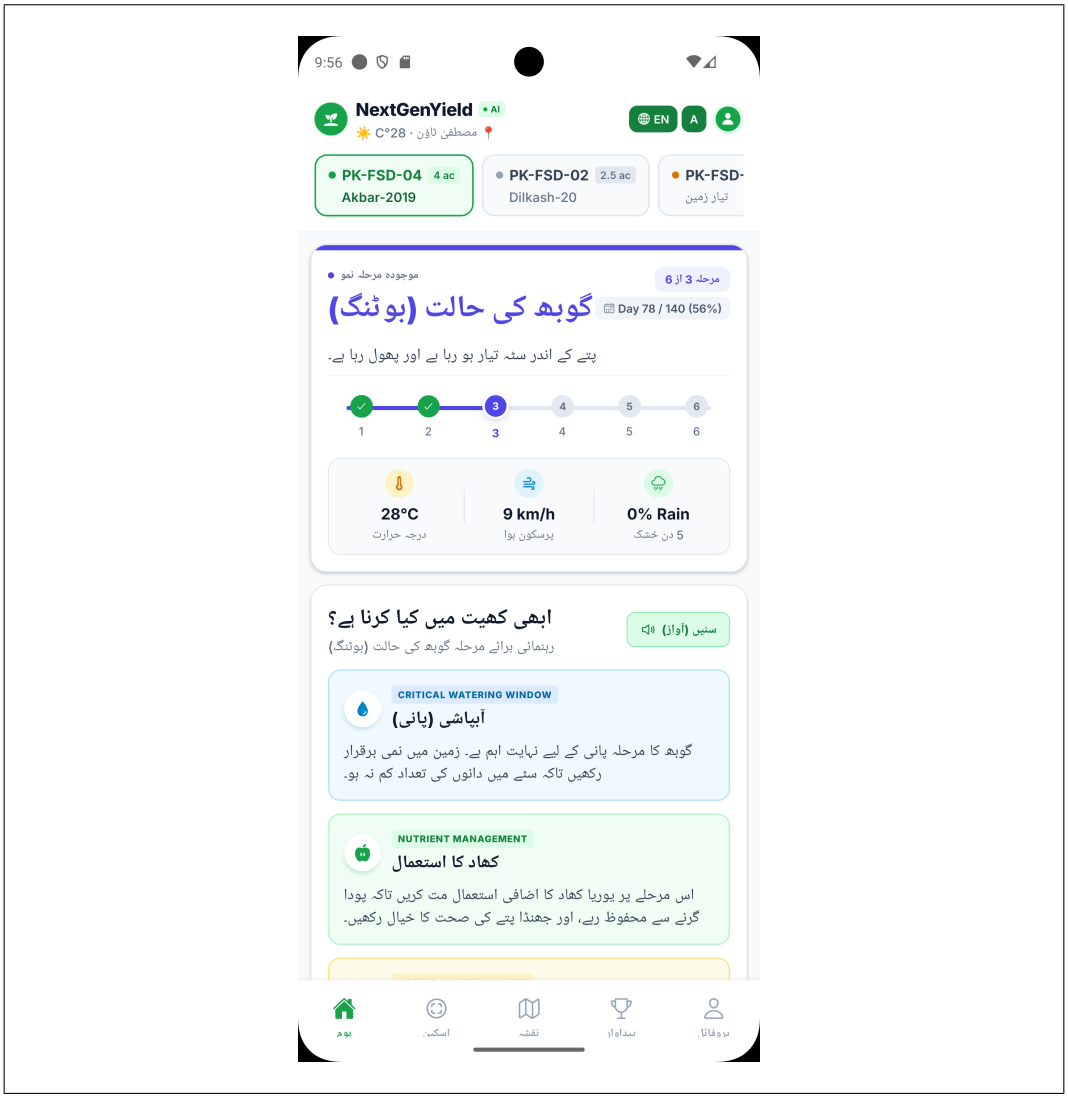


Figure 4.9: Farmer Accessibility and Spoken Audio Narration Settings

4.9 Database Design

The NextGenYield persistence layer is structured using a relational schema implemented in PostgreSQL with the PostGIS extension for geospatial polygon boundaries. The Entity-Relationship (ER) model and associated data dictionary tables are defined below.

4.9.1 ER Diagram

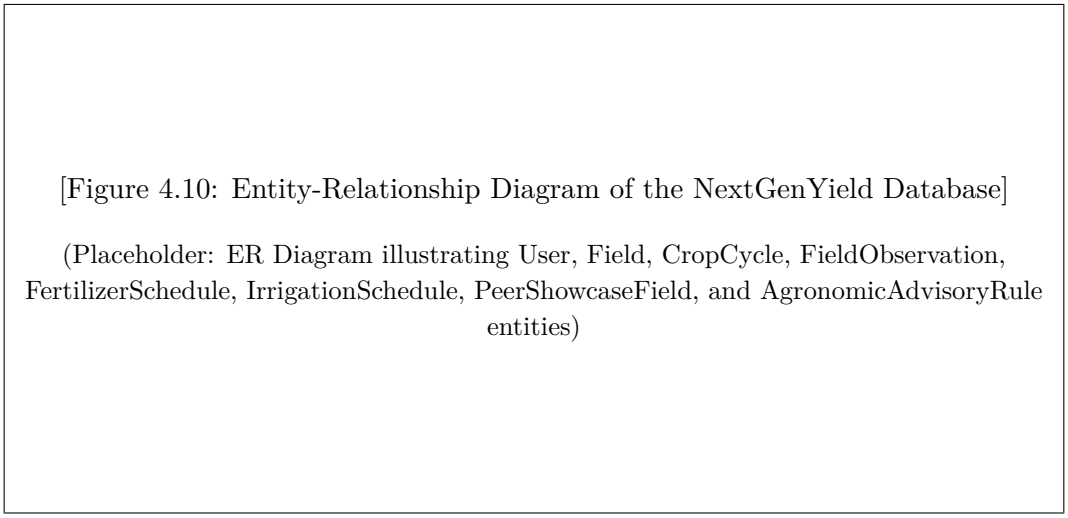


Figure 4.10: Entity-Relationship Diagram of the NextGenYield Database

4.9.2 Data Dictionary

The relational database architecture is defined by the following eight entity specifications:

Table 4.11: User Table Data Dictionary

Attribute	Data Type	Null	Relation	Rel Type	Description
UserID	Primary Key	No	-	-	Unique identifier for the user.
FullName	Varchar(60)	No	-	-	Full legal or preferred name of the user.
PhoneNumber	Varchar(15)	No	-	-	Primary contact number used for authentication.
PasswordHash	Varchar(255)	No	-	-	Bcrypt hashed security password/PIN.
Role	Varchar(20)	No	-	-	User role (‘farmer’, ‘landowner’, ‘stakeholder’).
District	Varchar(50)	No	-	-	Home administrative district (e.g., Faisalabad).
LanguagePref	Varchar(5)	No	-	-	Language code (‘ur’ for Urdu, ‘en’ for English).
CreatedAt	Timestamp	No	-	-	Record creation timestamp.

Table 4.12: Field Table Data Dictionary

Attribute	Data Type	Null	Relation	Rel Type	Description
FieldID	Primary Key	No	-	-	Unique identifier for the land parcel.
UserID	Foreign Key	No	User	Many-to-One	User who owns or manages the field.
FieldName	Varchar(50)	No	-	-	Friendly display name (e.g., North Canal Plot).
ShapeType	Varchar(15)	No	-	-	Geometry type ('square', 'rectangle', 'polygon', 'circle').
AreaAcres	Decimal(6,2)	No	-	-	Computed land area in standard acres.
AreaSqm	Decimal(10,2)	No	-	-	Computed land area in square meters.
BoundaryGeoJSON	JSONB	No	-	-	RFC 7946 GeoJSON polygon coordinates.
CenterLatitude	Decimal(9,6)	No	-	-	Geographic centroid latitude coordinate.
CenterLongitude	Decimal(9,6)	No	-	-	Geographic centroid longitude coordinate.
CurrentCycleID	Foreign Key	Yes	CropCycle	One-to-One	Active crop cycle pointer (null if fallow).
CreatedAt	Timestamp	No	-	-	Record creation timestamp.

Table 4.13: CropCycle Table Data Dictionary

Attribute	Data Type	Null	Relation	Rel Type	Description
CycleID	Primary Key	No	-	-	Unique identifier for the 140-day crop cycle.
FieldID	Foreign Key	No	Field	Many-to-One	Associated land parcel.
WheatVariety	Varchar(50)	No	-	-	Seed cultivar (Akbar-19, Dilkash-20, etc.).
SeedCompany	Varchar(60)	No	-	-	Certified seed provider or research agency.
FertilizerCompany	Varchar(60)	No	-	-	Partner manufacturer (FFC, Engro, Fatima).
SowingMethod	Varchar(30)	No	-	-	Planting method (Drill, Broadcast, Bed).
SowingDate	Date	No	-	-	Date seed was planted into moist soil.
TargetHarvestDate	Date	No	-	-	Projected 140-day maturity harvest date.
IsActive	Boolean	No	-	-	Flag indicating whether cycle is ongoing.
CreatedAt	Timestamp	No	-	-	Record creation timestamp.

Table 4.14: FieldObservation Table Data Dictionary

Attribute	Data Type	Null	Relation	Rel Type	Description
ObservationID	Primary Key	No	-	-	Unique identifier for visual scan record.
CycleID	Foreign Key	No	CropCycle	Many-to-One	Associated crop cycle.
ImageURI	Varchar(255)	No	-	-	Persistent file storage path for canopy photo.
StageIndex	Integer	No	-	-	Predicted stage index (0 to 5).
StageName	Varchar(30)	No	-	-	Predicted stage label (e.g., Booting Stage).
Confidence	Decimal(5,2)	No	-	-	FoMo4Wheat classification confidence score (%).
PitchAngleDeg	Decimal(4,1)	No	-	-	Measured camera pitch tilt angle ($45° - 80°$).
BlurScore	Decimal(6,2)	No	-	-	Laplacian edge variance sharpness metric.
DAS	Integer	No	-	-	Days After Sowing at time of capture.
CapturedAt	Timestamp	No	-	-	Exact timestamp of observation capture.

Table 4.15: FertilizerSchedule Table Data Dictionary

Attribute	Data Type	Null	Relation	Rel Type	Description
FertilizerID	Primary Key	No	-	-	Unique identifier for fertilizer split entry.
CycleID	Foreign Key	No	CropCycle	Many-to-One	Associated crop cycle.
SplitNumber	Integer	No	-	-	Sequential application split (1, 2, or 3).
NutrientType	Varchar(40)	No	-	-	Type (e.g., Basal DAP, 1st Top-dress Urea).
TargetDAS	Integer	No	-	-	Recommended Days After Sowing for application.
DosePerAcre	Varchar(50)	No	-	-	Recommended quantity (e.g., 1 Bag (50kg) / Acre).
IsApplied	Boolean	No	-	-	Flag indicating completed application.
AppliedAt	Timestamp	Yes	-	-	Timestamp of farmer confirmation.

Table 4.16: IrrigationSchedule Table Data Dictionary

Attribute	Data Type	Null	Relation	Rel Type	Description
IrrigationID	Primary Key	No	-	-	Unique identifier for irrigation event.
CycleID	Foreign Key	No	CropCycle	Many-to-One	Associated crop cycle.
EventName	Varchar(40)	No	-	-	Irrigation milestone (e.g., Rauni Pre-sowing, 1st CRI Irrigation).
WaterSource	Varchar(30)	No	-	-	Source (Canal Supply, Tube-well, Solar Pump).
TargetDAS	Integer	No	-	-	Target Days After Sowing (e.g., Day 22 for CRI).
IsCompleted	Boolean	No	-	-	Status indicating field watering completion.

Table 4.17: PeerShowcaseField Table Data Dictionary

Attribute	Data Type	Null	Relation	Rel Type	Description
ShowcaseID	Primary Key	No	-	-	Unique identifier for verified benchmark plot.
FieldCode	Varchar(20)	No	-	-	Identifier (e.g., PK-FSD-04 Benchmark).
District	Varchar(50)	No	-	-	Agricultural tehsil/district (e.g., Samundri, Faisalabad).
WheatVariety	Varchar(50)	No	-	-	Wheat cultivar sown.
SeedCompany	Varchar(60)	No	-	-	Certified seed provider.
FertilizerCompany	Varchar(60)	No	-	-	Fertilizer manufacturer utilized.
AreaAcres	Decimal(6,2)	No	-	-	Total parcel acreage.
ShapeType	Varchar(15)	No	-	-	Geometry type (‘rectangle’, ‘square’, ‘polygon’, ‘circle’).
YieldMaundsPerAcre	Decimal(5,1)	No	-	-	Verified harvest yield (e.g., 62.0 Maunds/Acre).
YieldKgSqm	Decimal(5,2)	No	-	-	Normalized harvest yield efficiency (kg/m^2).
SowingDate	Date	No	-	-	Sowing date of the verified cycle.
HarvestDate	Date	No	-	-	Harvesting date of the verified cycle.
CycleEventLog	JSON	No	-	-	Full chronological 140-day log of watering, weed filtering, and fertilizer splits.

Table 4.18: AgronomicAdvisoryRule Table Data Dictionary

Attribute	Data Type	Null	Relation	Rel Type	Description
RuleID	Primary Key	No	-	-	Unique identifier for deterministic rule.
StageIndex	Integer	No	-	-	Biological wheat growth stage index (0 to 5).
StageName	Varchar(30)	No	-	-	Growth stage identifier (Tillering, Booting, etc.).
SummaryTextEn	Text	No	-	-	English agronomic stage summary.
SummaryTextUr	Text	No	-	-	Urdu translated stage summary.
IrrigationAdviceEn	Text	No	-	-	English stage-specific watering directive.
IrrigationAdviceUr	Text	No	-	-	Urdu stage-specific watering directive.
NutrientAdviceEn	Text	No	-	-	English fertilizer timing guidance.
NutrientAdviceUr	Text	No	-	-	Urdu fertilizer timing guidance.
RiskAlerts	JSON	No	-	-	Active disease/weather hazard warnings array.
AudioScriptUr	Text	No	-	-	Phonetic Urdu audio synthesis script.

4.10 Risk Analysis

Risk assessment and management are foundational to ensuring NextGenYield's agronomic validity, operational resilience, and sustained adoption among smallholder farming communities. The principal risk dimensions and mitigation protocols are outlined below.

4.10.1 Technical and Environmental Vision Risks

- Risk Description: Outdoor agricultural photography is subject to severe environmental degradation, including wind-induced plant motion blur, intense direct solar glare ($> 60,000$ lux), shadow occlusions, and off-target horizontal camera angles pointing at weed foliage or lateral stems.
- Mitigation Strategy: NextGenYield implements on-device sensor fusion. Real-time accelerometer and gyroscope listeners enforce a strict downward pitch angle reticle ($45^\circ - 80^\circ$) before camera activation, and an edge Laplacian variance filter halts capture if motion blur exceeds thresholds (< 90). This guarantees that only crisp canopy imagery enters the vision pipeline.

4.10.2 Agronomic Safety and Misinformation Risks

- Risk Description: Automated agricultural advisory systems that deploy ungrounded Large Language Models (LLMs) are prone to hallucinations, generating fabricated chemical formulations, premature irrigation advice during high winds, or excessive nitrogen dosages that trigger vegetative lodging and ground-water leaching.
- Mitigation Strategy: NextGenYield eliminates probabilistic generative text models from the advisory pathway. Growth stage classifications are linked directly to deterministic, verified agronomic rule tables reviewed by certified agronomic research institutions.

4.10.3 User Adoption and Human Factors Risks

- Risk Description: Smallholder farmers in the Punjab agricultural belt frequently possess limited English literacy and smartphone experience. An overly complex, text-heavy interface risks complete abandonment or user error.
- Mitigation Strategy: The mobile user experience is designed with high visual contrast, intuitive icon metaphors, complete Urdu localization, and an integrated spoken Urdu audio synthesis player that narrates recommendations aloud at a single tap.

4.10.4 Rural Telecommunications and Connectivity Risks

- Risk Description: Remote agricultural parcels frequently suffer from weak, intermittent 2G/3G cellular signals, causing network timeouts and data transmission failures during field operations.
- Mitigation Strategy: The application incorporates offline-first local caching. Field boundaries, stage advisory guidelines, and photographic records are stored on-device in persistent local storage. Captured observations are queued and automatically synchronized once cellular reception is re-established.

4.10.5 Geospatial Demarcation and Land Discrepancy Risks

- Risk Description: Smallholder farms in Pakistan are subdivided into irregular, non-standard cadastral parcels (killas and acres) often bisected by irrigation canals, rendering rigid rectangular bounding tools ineffective.
- Mitigation Strategy: NextGenYield provides flexible multi-vertex polygon and centre-pivot circular boundary tools in addition to square and rectangular boxes, enabling farmers to accurately outline irregular parcels along irrigation watercourses and roads.

4.11 Conclusion

Chapter 4 has articulated the complete Software Requirement Specifications for NextGenYield. Beginning with a detailed feature breakdown and functional requirements for farmers, agricultural professionals, and backend systems, the chapter delineated non-functional performance and accessibility mandates. The nine formal use cases capture the end-to-end user journey—from GeoJSON boundary demarcation on satellite maps and single-entry cycle initialization to sensor-guided canopy photography, deterministic advisory generation, and peer benchmark replication. The graphical user interface architecture and relational database schema provide concrete implementation structures, while the risk analysis outlines rigorous mitigation strategies against vision degradation, agronomic hallucinations, and rural connectivity challenges. Together, these specifications lay an unyielding engineering foundation for precision wheat monitoring and decision support.

Chapter 5 Proposed Approach and Methodology

The proposed approach could be a Framework, Heuristic, Algorithm, Protocol, or Mathematical-model. For each chapter, provide a paragraph of introduction and in the end a paragraph of conclusions. Not the programming code but the algorithmic and procedural details especially related to the hidden/ backend algorithms that are not covered in the design.

NOTE: This chapter is optional for Development projects but compulsory for Research projects.

Chapter 6 High-Level and Low-Level Design

This following chapter specifies the high-level and low-level designs of NextGenYield. It includes the system overview, design considerations, system architecture, architectural strategies, class diagram, sequence diagrams, and the policies and tactics related to the project.

6.1 System Overview

In this section, we discuss the overall overview of the system. This encompasses all operational phases starting from land parcel demarcation, dataset ingestion, model training, edge-guided photography, and the deterministic advisory output delivered to the farmer.

6.1.1 System Design

The following describes the compositional structure of the system across its primary functional stages:

6.1.1.1 Field Demarcation and Cycle Initiation

The field demarcation module serves as the initial setup phase where farmers outline their land parcel boundaries directly on interactive satellite maps using four geometric shapes (Square, Rectangle, Polygon, and Circle). Once the boundary is established, the system computes the exact area in standard Acres and metric square meters (m^2). The farmer then initializes a 140-day wheat cycle by entering the certified seed variety (e.g., Akbar-19, Dilkash-20), seed provider, fertilizer company (FFC, Engro, Fatima), and sowing date.

6.1.1.2 Edge-Guided Canopy Photography

This module captures wheat canopy photos through the smartphone's camera while providing real-time quality assistance. Using hardware sensors (accelerometer and gyroscope), the interface guides the farmer to hold the phone at compliant oblique-to-nadir angles ($45^\circ - 80^\circ$). Simultaneously, an edge-based blur evaluator checks frame sharpness using Laplacian variance ($\sigma_L^2 \geq 100$) to reject blurry or shaky captures before image transmission.

6.1.1.3 Growth Stage Phenotyping

Once a compliant image is captured, it is forwarded to the AI inference service. The system utilizes the pre-trained FoMo4Wheat foundation model (ViT-B/14) to extract rich visual features from the canopy. A lightweight downstream classification head then categorizes the image into

one of six biological wheat growth stages: Tillering, Jointing, Booting, Anthesis, Grain Filling, or Maturity, along with an associated confidence score.

6.1.1.4 Advisory and Timeline Formulation

This stage translates classified growth stages into practical farming actions. Instead of using generative language models that risk hallucinating chemical dosages, the system deterministically maps the detected stage and cumulative Days After Sowing (DAS) to verified provincial agricultural extension protocols. It formulates clear guidance on irrigation timing, fertilizer split applications, and pathogen warnings (e.g., yellow stripe rust). These observations are chronologically logged into a temporal growth timeline, and an auditory engine vocalizes the recommendations in spoken Urdu.

6.1.1.5 Peer Benchmarking Showcase

This module highlights top-performing regional wheat fields normalized by surface area (kg/m^2 and Maunds/Acre). Farmers can inspect the complete 140-day crop management ledger of benchmark fields and adopt proven input schedules directly into their own field calendar with a single tap.

6.1.2 Data Collection and Ingestion

To train and validate the growth stage classification head, we utilize three complementary dataset sources:

- WGSP Benchmark Dataset: Multi-country wheat canopy imagery covering 263 varieties across all six biological growth stages.
- WGSP Smartphone Dataset: Handheld smartphone photography captured under variable ambient lighting, shadows, and natural field angles.
- Pakistani Field Realism Dataset: High-resolution field imagery collected from local Pakistani wheat trials (Khyber Pakhtunkhwa and Punjab), capturing indigenous soil textures, weed density, and local canopy characteristics.

6.1.3 Model Training

For growth stage classification, we attach a linear probing classification head on top of the pre-trained FoMo4Wheat Vision Transformer (ViT-B/14, 86.6M parameters). To preserve the extensive visual representations acquired during self-supervised pre-training on 2.5 million agri-

cultural images, the backbone weights remain completely frozen. The downstream head projects the combined 1536-dimensional feature vector (CLS token + pooled patch tokens) to the six biological stage classes using Cross-Entropy Loss with the AdamW optimizer.

6.2 Design Considerations

This section describes the assumptions and dependencies of the system, the general constraints, the goals and guidelines, and the development methods.

6.2.1 Assumptions and Dependencies

The project relies on the following operational assumptions:

- End-users have access to a commodity smartphone running Android 10.0+ or iOS 15.0+ with a functional camera, accelerometer, and GPS receiver.
- Adequate GPU computing resources (NVIDIA GPU with CUDA support) are available on the backend server for AI model inference.
- Farmers follow basic photography guidance by aiming the camera at the wheat canopy rather than bare soil or weeds.
- Standard OpenStreetMap or satellite tile services remain accessible for base layer GIS rendering.

6.2.2 General Constraints

NextGenYield operates within several environmental, technical, and domain constraints:

6.2.2.1 Hardware and Memory Constraint

Target mobile devices are often budget smartphones with limited RAM (2.0 GB to 3.0 GB). The client application must maintain an active memory footprint below 200 MB to prevent unexpected operating system termination.

6.2.2.2 Environmental Sunlight Constraint

Field inspections take place outdoors under intense ambient sunlight exceeding 50,000 lux. Interfaces must strictly use high-contrast light themes with contrast ratios exceeding $4.5 : 1$ (WCAG AA compliant) to preserve screen legibility.

6.2.2.3 Intermittent Connectivity Constraint

Rural agricultural regions frequently suffer from unstable 2G/3G cellular signals. The application must support offline local caching for captured observations, parcel boundaries, and advisory cards.

6.2.2.4 Input Quality and Angle Constraint

Photographs taken at improper angles (horizontal side-views) or degraded by wind motion blur compromise classification accuracy. The system enforces an onboard edge filter ($45^\circ - 80^\circ$ pitch angle and sharpness threshold ≥ 100) before allowing uploads.

6.2.2.5 Crop Scope Constraint

The system is explicitly designed for bread wheat (Triticum aestivum) cultivated in irrigated Indus Basin agricultural regimes. Other cereal or cash crops are excluded from the current operational release.

6.2.3 Goals and Guidelines

The overarching goals and guidelines shaping the software design include:

6.2.3.1 Goals

- The system will be user-friendly, requiring zero technical training for rural smallholder farmers.
- Recommendations will prioritize absolute agronomic safety through deterministic rule matching rather than probabilistic LLMs.
- The on-device edge filter will provide instant camera feedback (< 50 ms) to prevent wasted mobile data.
- Backend AI stage inference will complete in under 2.0 seconds over nominal cellular networks.

6.2.3.2 Guidelines

- Users should align the on-screen reticle until it turns green before tapping the shutter button.

- Users must register land parcel boundaries accurately using the provided zoomable satellite map.
- Recommended fertilizer doses and watering intervals should be applied during the specified phenological windows.

6.2.4 Development Methods

NextGenYield was developed using the Scrum agile development methodology. Development was partitioned into two-week sprints, each focusing on specific modular deliverables such as GIS polygon rendering, sensor fusion HUD integration, foundation model fine-tuning, and deterministic advisory database mapping. Periodic feedback from domain advisors and testing against local wheat trial images guided iterative refinements throughout the lifecycle.

6.3 System Architecture

The core responsibilities of NextGenYield are accurate phenological growth stage identification, real-time edge photography guidance, deterministic agronomic advisory formulation, and regional peer yield benchmarking.

To fulfill these responsibilities with high modularity and resilience, the system is organized around a Microservices and Layered Architecture as depicted in Figure 6.1. The mobile client communicates with backend services via secure RESTful APIs using standard JSON and multipart form-data protocols.

[Figure 6.1: High-Level System Architecture Diagram]

(Illustration showing Mobile Client Layer with Edge Sensor HUD → API Gateway → AI Inference Microservice [FoMo4Wheat ViT-B/14] → Deterministic Advisory & Timeline Engine → PostgreSQL/PostGIS Spatial Knowledge Repository → Peer Benchmarking Showcase)

Figure 6.1: NextGenYield High-Level Architecture Diagram

6.3.1 Subsystem Architecture

The system architecture is further decomposed into three specialized subsystems:

6.3.1.1 Edge Quality and Phenological Vision AI Subsystem

This subsystem is responsible for capturing compliant canopy photography and executing foundation model inference. It operates as a pipeline:

- Sensor Fusion Angle Check: Measures device pitch tilt using the onboard accelerometer ($45^\circ - 80^\circ$ acceptance window).
- Laplacian Blur Evaluator: Computes the spatial variance of the Laplacian kernel ($\sigma_L^2 \geq 100$) to detect motion blur.
- Image Preprocessing: Resizes raw canopy frames to 518×518 pixels and normalizes color channels with ImageNet standards.
- FoMo4Wheat Transformer Encoder: Extracts 1536-dimensional feature representations through the frozen ViT-B/14 backbone.
- Linear Classification Head: Outputs calibrated probabilities across the six biological wheat growth stages.

[Figure 6.2: Edge Quality and Vision AI Subsystem Architecture]

(Pipeline: Camera Feed → IMU Tilt & Laplacian Blur Filter → 518x518 Normalizer → Frozen FoMo4Wheat ViT-B/14 Backbone → Linear Probe Head → 6 WGSP Stages)

Figure 6.2: Edge Quality and Vision AI Subsystem Architecture

6.3.1.2 Geospatial Demarcation and GIS Subsystem

This subsystem manages spatial parcel boundaries and regional map visualization:

- Boundary Drawing Engine: Converts user touch interactions on satellite maps into standard geometric shapes (Square, Rectangle, Polygon, Circle).
- GeoJSON Serialization: Formats drawn coordinates into RFC 7946 GeoJSON `Polygon` objects.
- PostGIS Spatial Storage: Persists field perimeters using native `ST_Polygon` geometries indexed with spatial R-Trees.
- Geodesic Area Calculation: Computes real-time land surface area in Acres and square

meters (m^2).

[Figure 6.3: Geospatial Demarcation and GIS Subsystem Architecture]

(Pipeline: Map Touch Demarcation → Shape Geometry Validator → RFC 7946 GeoJSON Converter → PostGIS Spatial Database → Color-Coded Regional Map)

Figure 6.3: Geospatial Demarcation and GIS Subsystem Architecture

6.3.1.3 Deterministic Advisory and Growth Timeline Subsystem

This subsystem couples biological phenology with agronomic decision support:

- Timeline Tracker: Maintains an append-only chronological log of all field scans, tracking Days After Sowing (DAS) and enforcing monotonic progression ($0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5$).
- Deterministic Rule Matcher: Queries verified agricultural research tables based on the classified growth stage to retrieve precise fertilizer split doses and watering instructions.
- Audio Narration Synthesizer: Generates phonetic Urdu speech scripts for one-touch voice playback of all advisory recommendations.

[Figure 6.4: Deterministic Advisory and Timeline Subsystem Architecture]

(Pipeline: Stage Index & DAS Input → Monotonic Progression Check → Agronomic Rule Base Lookup → Localized Advisory Cards & Spoken Urdu Audio Output)

Figure 6.4: Deterministic Advisory and Timeline Subsystem Architecture

6.4 Architectural Strategies

The architectural strategies employed in designing NextGenYield are detailed below:

6.4.1 Programming Languages and Frameworks

The mobile client leverages TypeScript with the React Native / Expo framework. React Native provides native hardware sensor access (accelerometer, gyroscope, camera) and cross-platform consistency across Android and iOS while preserving a single codebase. For the backend AI services, Python is chosen due to its extensive ecosystem for machine learning and scientific computing (PyTorch, TorchVision). The FastAPI framework powers the REST API, delivering asynchronous performance, automatic OpenAPI documentation, and fast JSON request serialization.

6.4.2 Reuse of Existing Software Components

Rather than training a vision model from scratch on limited agricultural data, NextGenYield strategically reuses the pre-trained FoMo4Wheat foundation model (ViT-B/14). FoMo4Wheat was pre-trained on 2.5 million high-resolution wheat canopy images, offering rich visual priors that generalize across diverse varieties and soil conditions. Furthermore, PostGIS is leveraged for spatial polygon calculations, eliminating the need to write custom geospatial indexing algorithms.

6.4.3 User Interface Paradigms

The interface is engineered to be intuitive and accessible for rural users with varying digital literacy. A high-contrast light mode guarantees legibility under bright outdoor sunlight. Complex camera settings are replaced with a single self-guiding HUD reticle that turns green when the phone is held correctly. All agronomic advice supports instantaneous bilingual toggling (English and Urdu) and features a one-touch spoken audio button that narrates recommendations aloud.

6.4.4 Database Management

PostgreSQL coupled with the PostGIS extension serves as the primary database. PostgreSQL provides robust relational integrity for user accounts, crop cycles, and observation records, while PostGIS manages spatial polygon coordinates with spatial indexing. Additionally, PostgreSQL's native `JSONB` support enables efficient storage of chronological cycle event logs and disease alert arrays.

6.5 Domain Model/Class Diagram

The structural design of NextGenYield is represented by the class diagram illustrated in Figure 6.5.

[Figure 6.5: NextGenYield Class Diagram]

(Class Diagram illustrating User, Field, CropCycle, FieldObservation, FertilizerSchedule, IrrigationSchedule, AgronomicAdvisoryRule, PeerShowcaseField, EdgeGuardrailService, and AIInferenceService along with their attributes, methods, and relationship cardinalities)

Figure 6.5: NextGenYield Class Diagram

The core domain classes include:

- **`User`**: Stores user identity, authentication credentials, contact details, district, and preferred interface language.
- **`Field`**: Represents a permanent land parcel, containing shape geometry, GeoJSON polygon boundary, center coordinates, and calculated acreage.
- **`CropCycle`**: Encapsulates a seasonal 140-day wheat crop cycle, recording wheat variety, seed provider, fertilizer brand, sowing date, and active cycle status.
- **`FieldObservation`**: Logs individual canopy scans, including image URI, predicted growth stage index (0–5), confidence score, camera tilt angle, blur score, and Days After Sowing (DAS).
- **`FertilizerSchedule`**: Tracks stage-dependent fertilizer split applications (Basal DAP, Urea top-dressing, Zinc), target DAS, recommended dosage, and farmer completion status.
- **`IrrigationSchedule`**: Tracks critical watering events (Rauni pre-sowing, CRI, Booting), water source, and completion status.
- **`AgronomicAdvisoryRule`**: Stores deterministic guidance protocols, bilingual summaries, irrigation/nutrient recommendations, risk alerts, and Urdu audio scripts.

- **`PeerShowcaseField`**: Stores verified high-yield benchmark plots, normalized yield efficiency (kg/m^2), and adoptable 140-day management event logs.

6.6 Sequence Diagrams

The sequence diagrams illustrating key runtime interactions between users, the mobile application, backend gateways, AI services, and the database are presented below:

6.6.1 User Registration and Authentication

Figure 6.6 shows the sequence of steps for registering a new user and authenticating existing credentials.

[Figure 6.6: User Registration Sequence Diagram]

(User → Mobile UI → API Gateway → Database: Validates phone number uniqueness, hashes password, stores profile, and returns auth session token)

Figure 6.6: User Registration Sequence Diagram

6.6.2 Land Boundary Demarcation

Figure 6.7 illustrates how a farmer draws parcel boundaries on the satellite map, validates topology, and stores the GeoJSON polygon in PostGIS.

[Figure 6.7: Land Boundary Demarcation Sequence Diagram]

(User → Map Canvas → GeoJSON Converter → PostGIS Database: Computes surface area in acres/m^2 and stores parcel polygon)

Figure 6.7: Land Boundary Demarcation Sequence Diagram

6.6.3 Crop Cycle Initialization

Figure 6.8 depicts the initialization of an active 140-day wheat cycle for a registered field parcel.

[Figure 6.8: Crop Cycle Initialization Sequence Diagram]

(User → Mobile UI → Cycle Controller → Database: Associates seed variety, fertilizer partner, and sowing date with field, generating initial fertilizer and irrigation schedules)

Figure 6.8: Crop Cycle Initialization Sequence Diagram

6.6.4 Canopy Image Capture and Phenological Classification

Figure 6.9 shows the end-to-end flow from real-time edge angle and blur filtering to cloud-based FoMo4Wheat growth stage classification.

[Figure 6.9: Image Capture and AI Classification Sequence Diagram]

(Farmer → Camera HUD [IMU Angle + Blur Check] → Image Upload → FoMo4Wheat ViT-B/14 Inference Service → Returns Stage Index and Confidence %)

Figure 6.9: Canopy Image Capture and Stage Classification Sequence Diagram

6.6.5 Advisory Generation and Timeline Tracking

Figure 6.10 illustrates the generation of deterministic recommendations and logging the observation into the chronological timeline.

6.6.6 Peer Benchmark Playbook Adoption

Figure 6.11 depicts how a farmer inspects a high-yield peer field and adopts its management playbook into their active crop cycle.

[Figure 6.10: Advisory Generation and Timeline Tracking Sequence Diagram]

(AI Prediction → Timeline Engine [Monotonic Check] → Advisory Rule Lookup → Mobile UI: Renders bilingual advisory card and vocalizes Urdu narration)

Figure 6.10: Advisory Generation and Timeline Tracking Sequence Diagram

[Figure 6.11: Peer Benchmark Playbook Adoption Sequence Diagram]

(Farmer → Showcase View → Selects High-Yield Field → Adopt Playbook → System scales fertilizer splits to parcel acreage and updates calendar)

Figure 6.11: Peer Benchmark Playbook Adoption Sequence Diagram

6.7 Policies and Tactics

The following considerations govern the technical policies and tactical choices maintained throughout project implementation:

6.7.1 Tools

Model training, evaluation, and fine-tuning are performed using Python in PyTorch on NVIDIA GPU workstations. Visual Studio Code serves as the primary development environment for both the mobile client and backend microservices. Git and GitHub are utilized for collaborative version control and issue tracking.

6.7.2 Coding Guidelines

The mobile application adheres to strict TypeScript typing and modular component standards under React Native. Backend services follow standard Python PEP-8 coding conventions with Pydantic schemas for data validation. Code is organized into isolated functional directories to ensure maintainability.

6.7.3 Extensibility and Modularity

The architecture separates presentation, sensing, inference, and persistence into independent services. This modular structure allows future extensions—such as adding new crop types (e.g., rice or cotton) or incorporating multispectral satellite vegetation indices—without requiring major architectural overhauls.

6.7.4 User Interface Accessibility

The mobile interface prioritizes accessibility for rural smallholders through high-contrast visual tokens, intuitive icon navigation, bilingual English/Urdu toggling, and integrated audio synthesis that reads recommendations aloud in spoken Urdu.

6.7.5 Software Testing Strategy

A comprehensive testing strategy is employed, combining unit testing of individual services (e.g., GeoJSON geometry conversion and rule matching) with integration testing of the end-to-end camera-to-inference pipeline. Edge sensor fusion logic is validated through hardware simulation and on-ground field photography trials.

6.8 Conclusion

In conclusion, this chapter provides a clear and structured specification of NextGenYield's high-level and low-level designs. The system overview details the end-to-end journey from field demarcation to deterministic advisory delivery. Design considerations address real-world hardware, environmental sunlight, and connectivity constraints. The system and subsystem architectures establish modular pipelines for vision AI phenotyping, GIS demarcation, and rule-based decision support. Finally, the class diagram, sequence diagrams, and implementation policies ensure a robust engineering blueprint that directly supports the subsequent implementation phase.

Chapter 7 Implementation and Test Cases

For each chapter, provide a paragraph of introduction and at the end, a paragraph of conclusions. (Not the programming code but the algorithmic and procedural details especially related to the hidden/ backend algorithms that are not covered in the design)

7.1 Implementation

Whatever implementation that you have done so far, please elaborate here. Give clear details of the algorithms that were implemented along with the platform and the APIs which were used. NOTE: For FYP-1, this chapter can be changed to description of prototype developed.

7.1.1 Implementation of First Component/Algorithm

Write the implementation of the first component of your system here.

7.2 Test case Design and description

This section will be added in FYP-II. Summarize the common attributes of test cases. This may include input constraints that must be true for every input in the set of associated test cases, any shared environmental needs, any shared special procedural requirements, and any shared case dependencies. The following scheme is recommended for describing test cases in detail. The Sample test case is provided in table 7.1.

Table 7.1: Sample Test case No.1

<Software Component Name>			
<Reference>			
Test Case ID:	Reference Number	QA Test Engineer:	Name of personnel
Test case Version:	Version number	Reviewed By:	Testing Team lead
Test Date:	Date	Use Case Reference(s):	Relation to use cases
Revision History:	Refer to previous test case identity (if any)		
Objective:	Need and scope of the testing		
Product/Ver/ Module:	Refer to overall system being built and the place of this test case in it.		
Environment:	Necessary and desired properties of the test environment.		
Assumptions:	Assumptions that might affect the testing process.		
Pre-Requisite:	Necessary condition that needs to be fulfilled before the test case.		
Step No.	Execution description	Procedure result	
	Events being tested.	Mention software response.	
Comments:			
Passed Failed Not Executed			

7.3 Test Metrics

Summarize here the common ground of attributes of test case metrics as explained in Table 7.2.

Table 7.2: Sample Test case Matric.No.1

Metric	Purpose
Number of Test Cases	Total number of test cases that you have developed for your system.
Number of Test Cases Passed	The number of test cases that successfully passed
Number of Test Cases Failed	The number of test cases that failed
Test Case Defect Density	(No of test cases failed * 100) No of test cases executed
Test Case Effectiveness	No of defects detected using test cases *100 Total number of defects detected
Traceability Matrix	Traceability is the ability to determine that each the feature has a source in requirements and each requirement has a corresponding implemented feature.

7.3.1 Sample Test case Metric.No.2

Similar to the above table, add here for test metrics 2,3,4.

Chapter 8 User Manual

This chapter will be added in FYP-II. Design a complete user manual of your application and add that in this chapter. NOTE: This Chapter is optional for Research projects but compulsory for Development projects.

Chapter 9 Experimental Results and Discussion

This chapter will be added in FYP-II. Give proper analysis and discussion of experimental results (in plain English text) along with tables of results. For each chapter provide a paragraph of introduction and in the end a paragraph of conclusions.

NOTE: This Chapter is optional for Development projects but compulsory for Research projects.

Chapter 10 Conclusions

10.1 aa

10.1.1 dfs

10.1.1.1 dsd

This chapter is mandatory. Give conclusions and summary of the work done. What were your findings and what were the results? Discuss in detail whether the scope of your project was entirely covered or not and whether the objectives of the project were met or not. What challenges did you face and what has been left out and why.

Sum up all the conclusions of all the chapters here to make a final conclusion chapter. Do not repeat any text, just summarize it in different words. Give recommendations for future work also.

For FYP-1 it is mandatory to list down a plan of the work to be done for FYP-2.

Bibliography

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[8] B. Han, C. Zhu, D. Han, et al., "FoMo4Wheat: Toward reliable crop vision foundation mod-

els with globally curated data." arXiv preprint arXiv:2509.06907, 2025. Available: https://arxiv.org/abs/2509.06907; code: https://github.com/PheniX-Lab/FoMo4Wheat.

Chapter A First Appendix if Required

If you want to add appendices then make sure you provide a proper title. Add different appendices by using the chapter command as shown for this one. They will automatically be added to the table of contents.

A.1 References

See how .bib file is prepared. The bibliography would be automatically added by Latex. So here is how you cite papers (you'll need to view the .bib file to understand). [?] is the correct way to make one citation and [? ?] to make multiple citations. You can see various reference styles for manuals [?], conferences [?], websites [?] and journal articles [?]. Also, note that all citations appear in the order in which they appear within text. They are not sorted according to author names.

A.2 Equations

Make sure you use the math environment for writing equations. Again, make sure you do not hard code equation numbers. Use label and ref commands of Latex to number them and reference them. Here is an example of (A.1)

$$E = mc^2 \tag{A.1}$$

Note, how the equation is referenced by its equation number within brackets.

A.3 Figures

To insert a figure, you can use the following code as template. Here the figure is referenced using ref command of Latex. So Figure A.1 is the FAST logo. DO NOT HARD CODE FIGURE NUMBERS. Let Latex manage them for you.



Figure A.1: Fast Logo is the Caption.

A.4 Tables

Note how tables can be added in Latex. Also, make sure you use label and ref commands to define and use table numbers. You can use Table A.1 as a template to make tables, add captions to them and reference them. DO NOT HARD CODE TABLE NUMBERS. Let Latex manage their numbering for you.

If you are having problems in managing Latex tables, you may use Lyx https://www.lyx.org/.

Table A.1: Give a Caption to the Table.

No.	Column 1	Column 2
1.	Some data	More data
2.	Some values	More values

A.5 Pseudo Code

When you need to explain an existing algorithm of an algorithm you devised, then you should use pseudo code notation. Given below, algorithm 1, is an example. Note how the numbering is managed by the Latex itself.

Data: this text
Result: how to write algorithm with LaTeX2e
initialization;
while not at end of this document do
    read current;
    if understand then
        go to next section;
        current section becomes this one;
    else
        go back to the beginning of current section;
    end
end

Algorithm 1: How to write algorithms

A.6 Code of Programming Languages

If you need to add C++, Java or any other programming language's code, then you can use "listings" package. Following is an example. Normally, the code is only added in the appendices

to avoid clutter in the document.

```
#include <stdio.h>
#define N 10
/* Block
 * comment */

int main()
{
    int i;

    // Line comment.
    puts("Hello world!");

    for (i = 0; i < N; i++)
    {
        puts("LaTeX is also great for programmers!");
    }

    return 0;
}
```

A.7 Recommendations for LaTeX

Here we are providing you with only two Latex source files. main.tex and fypbib.bib file. You can also start a project at overleaf.com and upload the zip file of this template there and share the Latex source between your group and supervisor. This way all of you can work on the Latex files together online.